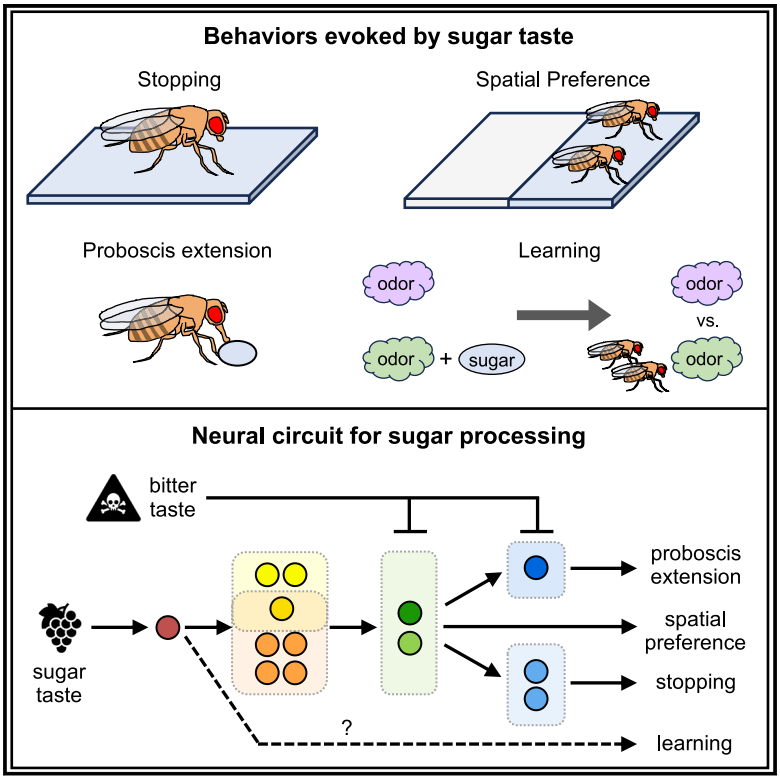
Cell Reports

Report

Overlap and divergence of neural circuits mediating distinct behavioral responses to sugar

Graphical abstract



Highlights

- Circuits for distinct sugar-evoked behaviors diverge at the second and third layers
- Sugar interneurons likely act synergistically to promote circuit activity
- Bitter input suppresses the sugar circuit through multiple pathways
- Modulatory factors act at different sites in the sugar circuit to regulate behavior

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In brief

Sensory stimuli, such as the taste of sugar, can evoke diverse behavioral responses. Jacobs et al. show that circuits for different innate responses to sugar in *Drosophila* partially overlap and diverge at the second and third layers. Modulatory factors act at different sites in the sugar circuit to influence behavior.

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Report

Overlap and divergence of neural circuits mediating distinct behavioral responses to sugar

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SUMMARY

How do neural circuits coordinate multiple behavioral responses to a single sensory cue? Here, we investigate how sweet taste drives appetitive behaviors in *Drosophila*, including feeding, locomotor suppression, spatial preference, and associative learning. We find that neural circuits mediating different innate responses to sugar are partially overlapping and diverge at the second and third layers. Connectomic analyses reveal distinct subcircuits that mediate different behaviors. Connectome-based simulations of neuronal activity predict that second-order sugar neurons act synergistically to promote downstream activity and that bitter input overrides the sugar circuit through multiple pathways acting at third- and fourth-order neurons. Consistent with the latter prediction, optogenetic experiments suggest that bitter input inhibits third- and fourth-order sugar neurons to override the sugar pathway, whereas hunger and diet act earlier in the circuit to modulate behavior. Together, these studies provide insight into how circuits are organized to drive diverse behavioral responses to a single stimulus.

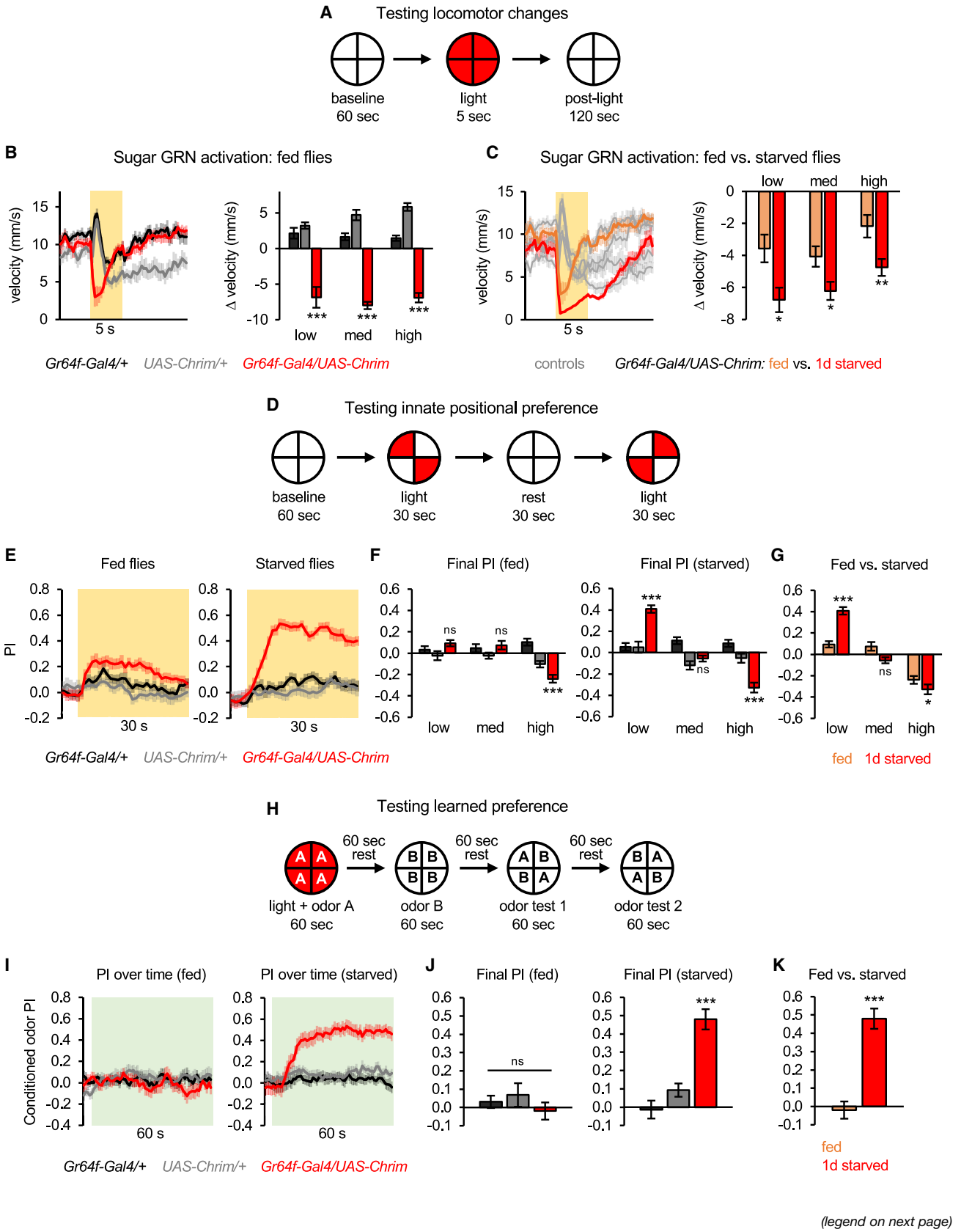
INTRODUCTION

A single sensory cue can elicit diverse behavioral responses—either simultaneously, sequentially, or on different timescales. For example, feeling that you have stepped on a sharp object may cause you to simultaneously wince and withdraw your leg, followed by inspecting your foot and forming a memory that alters future behavior, such as choosing not to walk barefoot in that area again. At what point does a sensory circuit diverge into different pathways that drive different behaviors? Where and how this divergence occurs has implications for how much sensory processing occurs prior to motor activation, how specific this processing is for different behaviors, and whether modulatory factors act globally or selectively to modulate sensory-evoked behaviors.

Studies in the fruit fly *Drosophila melanogaster*, where the architecture and function of neural circuits can be mapped at the level of single cells, are well suited to provide insight into the organization of sensory-motor circuits. Here, we investigate how sweet taste drives multiple appetitive behaviors in *Drosophila*. Flies detect sugar through sweet-sensing gustatory receptor neurons (GRNs) in the legs and proboscis, the fly's feeding organ.[1] Studies of sugar responses have primarily focused on food consumption or proboscis extension,[1–10] which represents the initiation of feeding, but the taste of sugar also influences other behaviors: it suppresses walking (to enable flies to feed),[11,12] causes preference for sugar-containing areas,[12,13] drives appetitive learning,[14,15] influences egg-laying site selection,[16,17] and modulates courtship and aggression.[18,19] Feeding responses to sugar can be modulated by internal and external cues, as feeding is enhanced by starvation[3,20,21] and suppressed by bitter taste.[4,22] Flies on a high-sugar diet show reduced proboscis extension to sugar[7,10,23,24] but increase their sugar consumption over several days.[10]

In this study, we address three major questions. First, are different sugar-evoked behaviors modulated in similar or different ways by factors such as hunger, diet, and bitter taste? Second, at what point does the sugar circuit diverge into different behavioral pathways? Third, where do modulatory inputs act on the sugar circuit to alter behavior? We first determined that different sugar-evoked behaviors are coordinately enhanced or suppressed by modulatory factors. We next found that circuits mediating different innate responses to sugar are partially overlapping and diverge at the second and third layers. Connectomic analyses, simulations of neuronal activity, and optogenetic manipulations revealed how sugar interneurons interact and how modulatory factors act on the sugar circuit, suggesting that different factors act at different sites. Together, these studies provide insight into how neural circuits coordinate diverse behavioral responses to a single stimulus.

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RESULTS

Activating sugar GRNs elicits locomotor suppression, positional preference, and appetitive learning

We established an optogenetic paradigm to study behaviors elicited by sweet taste. We used Chrimson[25] to optogenetically activate sugar GRNs using *Gr64f-Gal4*,[2] and we focused on three behaviors elicited by sugar: locomotor suppression,[11,12] positional preference,[12,13] and associative learning.[14,15] Activating sugar GRNs strongly suppressed forward velocity and turning (Figures 1A–1C and S1A), as previously observed.[26,27] In fed flies, this locomotor suppression was transient, whereas locomotor suppression was enhanced and prolonged in flies that were starved for 1 day (Figure 1C). We then tested whether flies prefer residing in areas where they receive sugar GRN activation (Figure 1D). Activating sugar GRNs at low light intensity elicited strong positional preference in 1-day-starved flies but only a weak, non-significant effect in fed flies (Figures 1E–1G). Unexpectedly, the attraction diminished at medium light intensity and shifted to aversion at high intensity (Figures 1F and 1G). Finally, we tested associative learning by quantifying flies' preference for an odor that was paired with sugar GRN activation (Figure 1H). We observed learned preference in starved flies but not fed flies (Figures 1I–1K), consistent with studies using sugar,[28,29] and this preference occurred with multiple light intensities and pairing protocols (Figure S1B). Together, these results show that the activation of sugar GRNs drives behaviors evoked by natural sugar, and all of these behaviors are enhanced by starvation. We verified these results using additional light stimulation parameters (Figures S1C and S1D) and a different Gal4 line, *Gr5a-Gal4*[2] (Figures S1E–S1G).

Given that activating sugar GRNs caused the opposite effects from activating bitter GRNs,[30] we wondered how hunger modulates the same behaviors in response to opposing tastes. We compared the effects of bitter GRN activation in fed and starved flies and found that hunger enhanced aversive learning but did not affect locomotor responses or positional aversion, even when flies were starved for 2 days (Figures S1H and S1I). These results show that hunger can act selectively to modulate specific taste pathways and behavioral responses, with a stronger and broader effect on sugar-evoked behaviors.

Behavioral responses to sweet taste are modulated by bitter input and a high-sugar diet

We asked how behaviors evoked by sugar-sensing GRNs are affected by simultaneously activating bitter-sensing GRNs. In all three behavioral assays, bitter input dominated over sugar input to drive aversive responses (Figures 2A–2G). However, we noted that flies with sugar and bitter GRN co-activation showed very brief locomotor suppression before increasing their speed (Figures 2A and 2B), suggesting that sugar and bitter inputs are integrated differently across time.

We then asked how sugar-evoked behaviors are affected by raising flies on a high-sugar diet. The high-sugar diet blunted all behavioral responses to sugar GRN activation, with flies exhibiting more transient locomotor suppression (Figures 2H and S2) and failing to show significant innate or learned preference (Figures 2I and 2J). Taken along with previous studies testing proboscis extension,[10,24] our results suggest that a high-sugar diet globally dampens short-term behavioral responses to sugar. Because we used optogenetic stimulation of sugar GRNs, which bypasses activation of sugar receptors, our results also demonstrate that a high-sugar diet decreases sugar sensitivity independently of any effect on sugar receptors. Together, our results show that modulatory factors such as hunger, bitter input, or diet coordinately enhance or suppress multiple sugar-evoked behaviors.

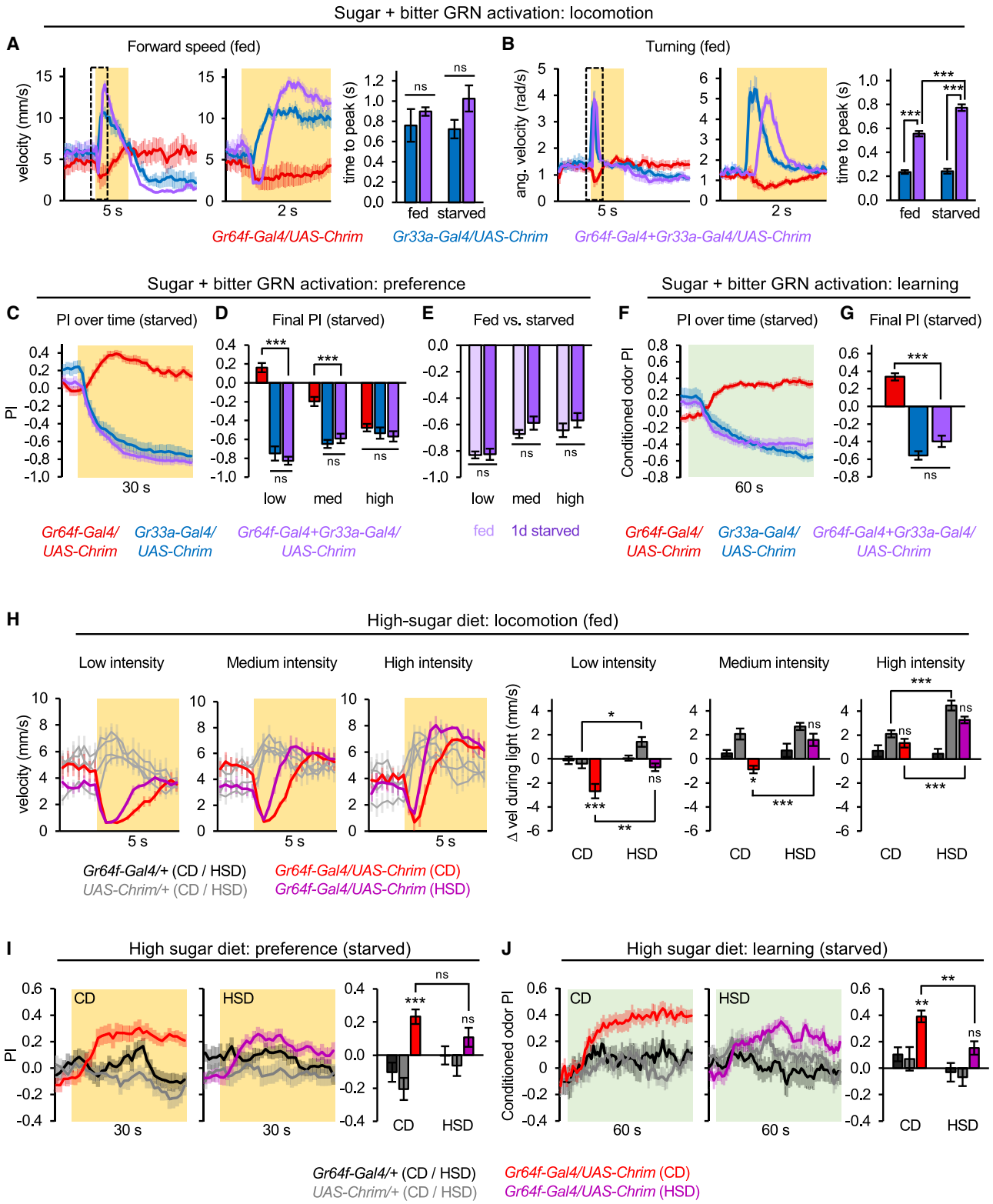
Overlap and divergence of pathways for different sugar-evoked behaviors

We next investigated how the sugar circuit diverges into different behavioral pathways. A recently identified circuit for proboscis extension to sugar includes three layers of interneurons residing in the subesophageal zone (SEZ; Figure 3A), and each interneuron causes proboscis extension when activated.[8] We tested whether activating each interneuron also drives other sugar-evoked behaviors in starved flies. We targeted each of the 10 interneuron types using previously characterized, highly specific split-Gal4 lines[8,31] whose expression patterns we verified (Figure S3A). We also confirmed that activation of each interneuron elicits proboscis extension, as previously demonstrated,[8] with the exception of one line (Figure S3B).

Individually activating five of the 10 interneuron types caused locomotor suppression (Figures 3B and S4A), including four of the seven second-order neurons (Rattle, FMIn, G2N-1, Zorro) and one of the two third-order neurons (Fdg). Four of the 10 interneuron types (Rattle, FMIn, Fdg, Bract) caused positional preference (Figures 3C and S4B), although this effect was only observed with one of the two split-Gal4 lines for Bract. None of the 10 interneurons caused significant learned preference (Figure S4C; see Figure S4D for correlations across behaviors). These results (Figure 4D) demonstrate that some second- and third-order neurons can drive several types of behavioral responses, implying that they connect to multiple behavioral pathways, whereas the fourth-order neuron only affected proboscis extension. Thus, circuits mediating different innate responses to sugar likely diverge at the level of second- and third-order neurons. Circuits for innate versus learned behaviors may diverge earlier—at the first synapse—given that none of the interneurons elicited associative learning, but we cannot rule out the possibility

Figure 1. Activating sugar GRNs elicits locomotor suppression, positional preference, and appetitive learning

Effects of sugar GRN activation on locomotion (A–C, n = 11 or 12 trials), positional preference (D–G; n = 22–24 trials), or learning (H–K; n = 16–24 trials) in fed or starved flies. For locomotion and preference, light stimulation was delivered at three intensities (low, medium, high). For positional preference and learning, behavior is quantified by the preference index (PI). In all figures, yellow shading indicates light on and green shading indicates odors on. Traces in (B), (C), and (E) show behavior with low-intensity light. Bar graphs for locomotion show change in velocity at light onset (B) or during the entire light period (C). See also Figure S1. For all figures: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; ns, not significant ($p > 0.05$). Unless otherwise specified, error bars represent standard error of the mean. See STAR Methods for statistical tests used.



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that we might observe learning effects with different protocols or if multiple interneurons were activated simultaneously.

Our results indicate that some sugar interneurons are sufficient to drive behavior, but are they necessary? To test this, we optogenetically activated sugar GRNs while silencing individual downstream neurons with Kir2.1.[32] Silencing Rattle, FMIn, or Fdg did not impair locomotor suppression elicited by sugar GRNs (Figure S4E). While we cannot rule out the possibility that the silencing was not effective (although we have observed strong phenotypes when expressing the same Kir2.1 transgene in other neurons[33]), these results suggest that individual second- and third-order neurons are sufficient but not necessary for locomotor suppression.

We then asked whether the behavioral effects driven by second- or third-order interneurons are modulated by hunger state or a high-sugar diet, as observed with sugar GRNs (Figures 1 and 2). Observing modulation would imply that these factors alter neural processing within or downstream of the second- or third-order neurons in addition to their known effects on sugar GRNs.[10,20,23,24] Locomotor suppression elicited by activating Rattle, FMIn, or Fdg was not modulated by either hunger or a high-sugar diet (Figures 3E and 3F), implying that hunger and diet act early in the circuit—upstream of second- or third-order neurons, and potentially within sensory cells—to modulate the effect of sugar on locomotion.

Connectivity analysis of sugar pathway interneurons

Our results suggest that second-order sugar neurons are functionally diverse. To gain insight into whether their functional diversity relates to their downstream connectivity, we identified their postsynaptic targets using a whole-brain connectome.[34,35] These analyses revealed two separate groups of second-order neurons that share many of the same targets: one group comprising Phantom and Usnea, and the other group comprising Zorro, Rattle, FMIn, and G2N-1 (Figure 3G; see Figure S4F for an analysis considering connection strength). Because Phantom and Usnea are predicted to be inhibitory and the other second-order neurons are predicted to be excitatory,[8] these analyses reveal a separation between inhibitory and excitatory subcircuits within the sugar pathway. Clavicle is more closely linked to the excitatory subcircuit but shares more targets with Phantom and Usnea than other excitatory second-order neurons, representing an intermediate role.

The four second-order neurons that caused locomotor suppression (Zorro, Rattle, FMIn, and G2N-1) are the four neurons that form the core excitatory subcircuit. All four of these neurons project to Fdg, which also suppressed locomotion, suggesting that Fdg represents a critical node for this behavior (Figure 3H). Fdg projects to Foxglove and Bluebell, two recently identified fourth-order neurons that cause stopping,[36] with Fdg's input to Foxglove being exceptionally strong (>100 synapses). Foxglove and Bluebell also receive direct input from FMIn and Rattle, but not other second-order neurons, which may explain why FMIn and Rattle elicited much stronger locomotor suppression than Zorro or G2N-1. In experiments described later, we confirmed that activation of Foxglove and Bluebell elicits locomotor suppression (see Figure 5C) but does not affect proboscis extension (Figure S3B), further demonstrating that circuits for different sugar-evoked behaviors diverge by the fourth layer.

Computational simulations suggest that second-order neurons have synergistic roles

To predict how connectivity patterns contribute to activity within the sugar circuit, we simulated brain-wide neuronal activity using a connectome-based model developed by Shiu et al.,[37] which relies on neurotransmitter predictions for each neuron.[38] We first used the model to ask how second-order neurons influence downstream activity in the sugar circuit. We bilaterally activated second-order neurons individually or in combination and analyzed the activity of downstream neurons, focusing on Fdg, Foxglove, and Bluebell (Figure 4A). Although four second-order neurons directly synapse onto Fdg, only Rattle was capable of activating Fdg on its own, and this activation was relatively weak. Activating Rattle in combination with G2N-1 or Zorro caused greater Fdg activity than Rattle alone, and activating all three neurons further increased Fdg activity. We observed similar effects on Foxglove and Bluebell, suggesting that the four second-order neurons that cause locomotor suppression act synergistically to activate the third- and fourth-order neurons that cause stopping. Simulations also showed that activation of Rattle or FMIn, but not other second-order neurons, suppresses activity in the walking pathway[36] (Figure S5A), consistent with our experimental data showing that these two second-order neurons have the strongest roles in locomotor suppression (Figure 3B).

Next, we used the model to activate sugar GRNs while bilaterally silencing second-order neurons. The activities of Fdg, Foxglove, and Bluebell were unaffected by individually silencing most second-order neurons, whereas simultaneously silencing multiple second-order neurons had stronger effects (Figure 4B). These results support the idea that second-order neurons act together to promote downstream activity and individual neurons are largely redundant, consistent with the negative results observed with experimental silencing (Figure S4E).

Computational simulations suggest that bitter input suppresses the sugar circuit via multiple pathways acting at third- and fourth-order neurons

The observation that bitter GRN activation completely overrides most sugar-evoked behaviors (Figures 2A–2G) suggests that

Figure 2. Behaviors elicited by sugar GRNs are modulated by bitter input or a high-sugar diet

(A–G) Effects of co-activating sugar and bitter GRNs on locomotion (A and B; $n = 9$ or 10 trials), preference (C–E; $n = 18$–20 trials), or learning (F and G; $n = 20$ trials). Traces in (A)–(C) show behavior with low-intensity light. In (A) and (B), right traces show zoomed-in behavior around light onset (boxes in left graphs) and bar graphs quantify the time to peak velocity after light onset.

(H–J) Effects of activating sugar GRNs on locomotion (H; $n = 7$–10 trials), preference (I; $n = 14$–20 trials), or learning (J; $n = 10$–18 trials) in flies fed on a high-sugar diet (HSD) or control diet (CD). Locomotor traces are zoomed in relative to other figures to show differences during the light stimulation period. Bar graphs show change in velocity during the light period. See also Figure S2.

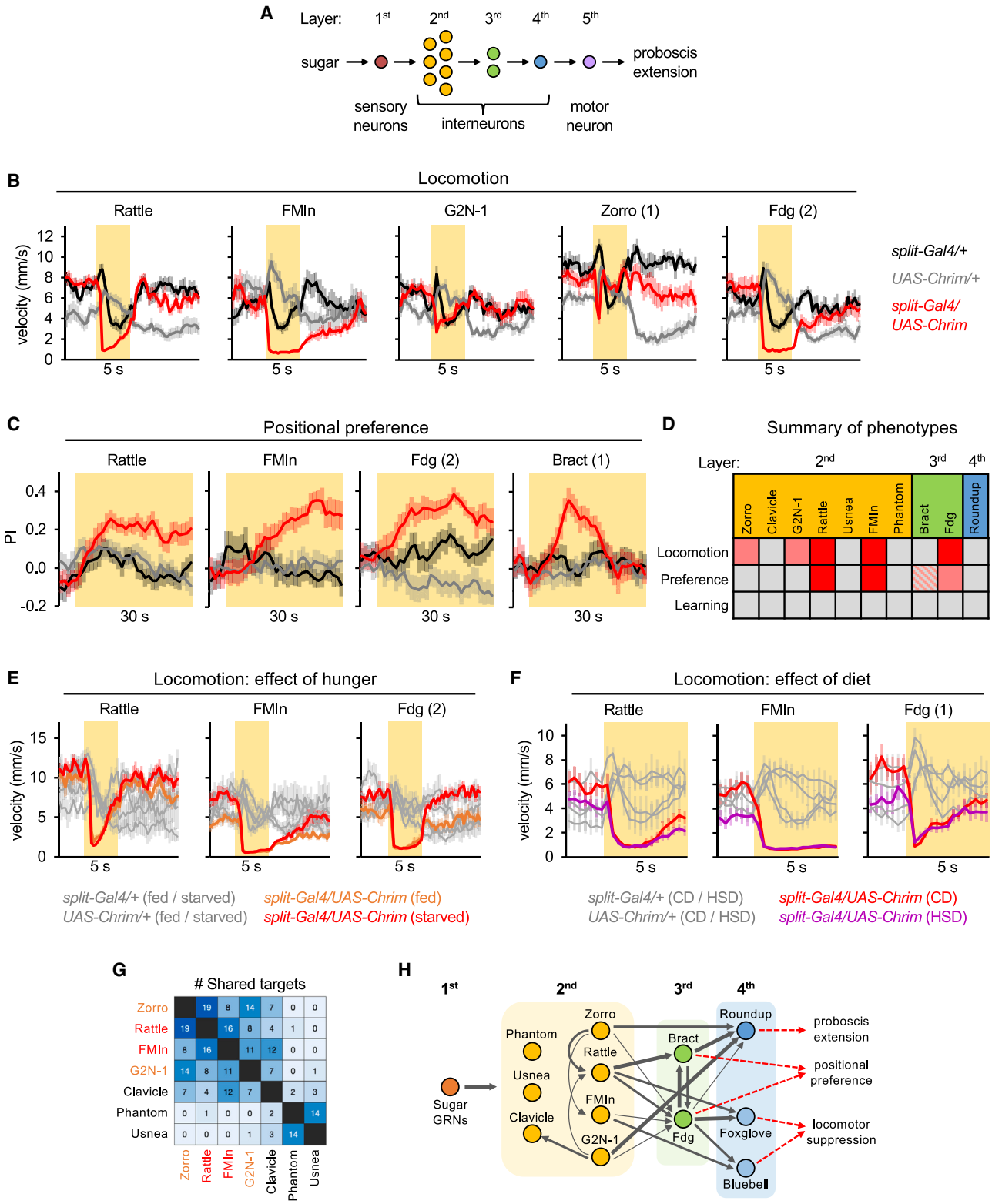


Figure 3. Overlap and divergence of pathways for different sugar-evoked behaviors

(A) Schematic of the neural circuit that drives proboscis extension to sugar.[8]

(B and C) Phenotypes of sugar interneurons that caused locomotor suppression (B; *n* = 7–12 trials) or preference (C; 16–24 trials).

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bitter input suppresses sugar-evoked activity in addition to driving competing behavioral pathways. To predict where this suppression occurs, we simulated co-activation of sugar and bitter GRNs at the same intensity and compared the results with sugar GRN activation alone. We observed a trimodal distribution corresponding to sugar-responsive neurons that were strongly suppressed, weakly suppressed, or not suppressed by bitter input (Figures 4C and S5B). None of the seven known second-order sugar neurons were suppressed by bitter input, whereas all known third- and fourth-order neurons showed some degree of suppression (Figure 4D). Neurons in the proboscis extension pathway (Roundup, MN9) were more strongly suppressed than those in the locomotor stopping pathway (Foxglove, Bluebell).

We then sought to identify candidate neurons that mediate bitter suppression of the sugar pathway. From the list of neurons that responded to simulated activation of bitter GRNs,[37] we focused on three types that have been anatomically defined: Scapula (previously identified as a second-order bitter neuron),[8] Mime, and Marge.[31] First, we simulated co-activation of each neuronal type along with sugar GRNs to examine suppression of sugar-evoked activity (Figure 4E). None of the three types of bitter neurons strongly suppressed activity in any known second-order sugar neurons, similar to bitter GRN activation (Figures 4F and S5C). Mime strongly suppressed the activity of all known third-, fourth-, and fifth-order sugar neurons, whereas Scapula strongly suppressed only neurons in the proboscis extension pathway (Roundup and MN9) and Marge strongly suppressed only Bract and Bluebell (Figures 4F and S5C).

We also asked whether these downstream bitter neurons are necessary for bitter suppression of the sugar pathway by silencing their activity while co-activating sugar and bitter GRNs. Silencing Scapula globally reduced bitter suppression, whereas silencing Mime or Marge had weaker effects (Figures 4G, 4H, and S5D). Bitter suppression of Roundup and MN9 was abolished by silencing Scapula but not Mime or Marge, whereas silencing Mime or Marge reduced bitter suppression of Bract and Bluebell (Figure 4I). Together, these results suggest that downstream bitter neurons play distinct roles in suppressing the sugar circuit, with some neurons selectively suppressing subcircuits for particular behaviors (Figure 4J).

Co-activation experiments suggest that bitter input suppresses third- and fourth-order sugar neurons

Our simulations predict that bitter inhibition of the sugar circuit occurs at the level of third- and fourth-order neurons (Figures 4D, 4F, and 4J). To test this prediction experimentally, we co-activated bitter GRNs with sugar interneurons. Co-activating bitter GRNs with sugar interneurons residing at or upstream of the inhibitory site(s) should allow bitter to override sugar-evoked behavior, whereas co-activation with sugar neurons downstream of the inhibitory site(s) should bypass bitter inhibition (Figure 5A). In the locomotor assay, the effect of bitter GRNs dominated when they were co-activated with second- or third-order sugar neurons (Figure 5B), but not fourth-order neurons (Figure 5C), suggesting that bitter suppression at the third layer is critical for overriding the sugar circuit to enhance locomotion. Results with positional preference assays were not easily interpretable (Figure S6A).

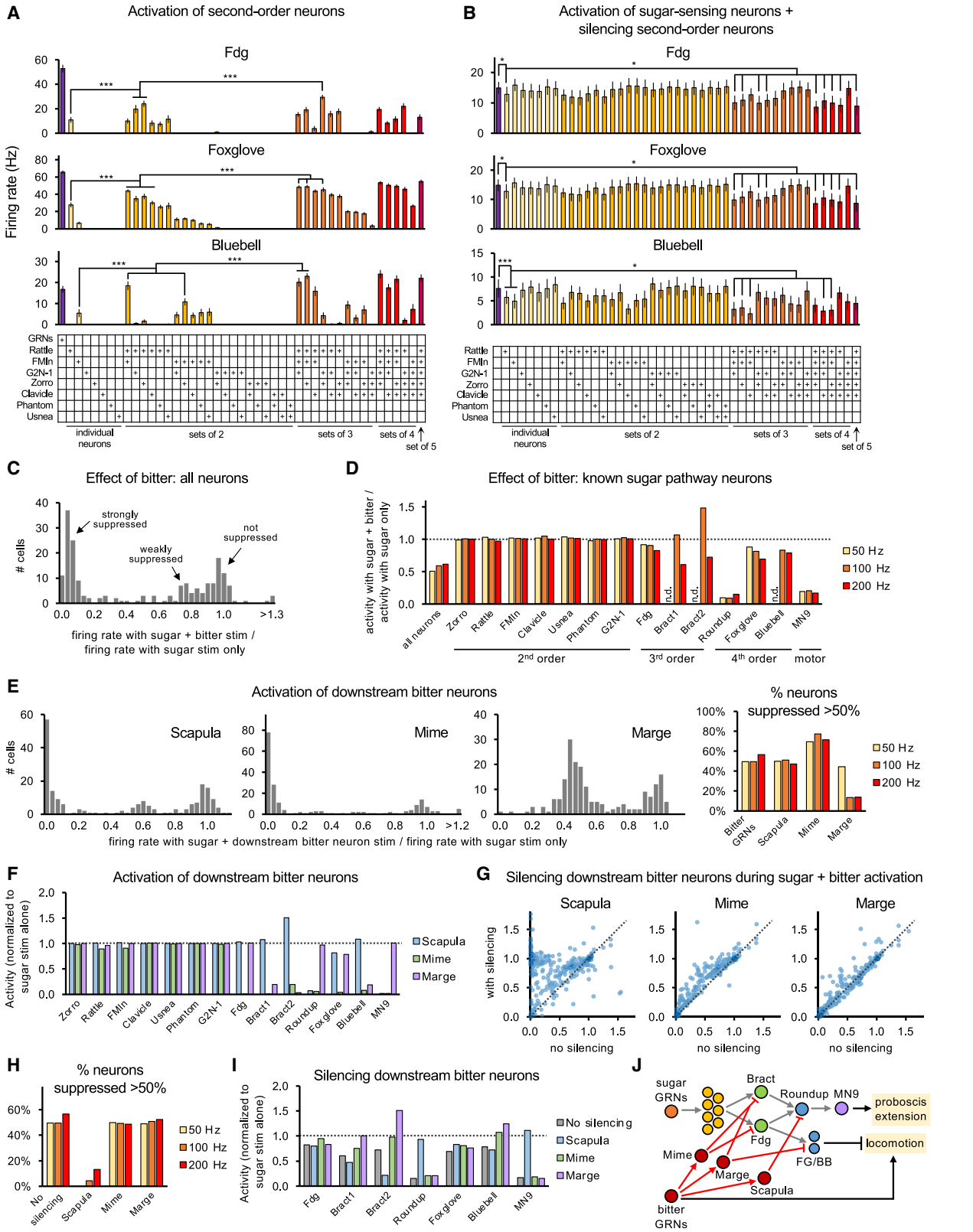
We also tested proboscis extension and found that co-activation of sugar and bitter GRNs caused proboscis extension in a small subset of flies, representing an intermediate phenotype between activation of sugar and bitter GRNs alone (Figure 5D). Bitter suppression of feeding responses has been linked to inhibition of the sugar circuit at multiple levels, including sugar GRNs[4,22] and the fourth-order neuron Roundup.[8] If inhibition of sugar GRNs plays the stronger role, then co-activation of bitter and sugar GRNs would evoke less proboscis extension than co-activation of bitter GRNs with second- or third-order sugar neurons, as the latter manipulation would bypass sugar GRN inhibition. We found that co-activation of bitter GRNs with second- or third-order neurons resulted in intermediate levels of proboscis extension, similar to co-activation of bitter and sugar GRNs (Figure 5D), indicating that none of these sugar interneurons could override the effect of bitter. Thus, downstream bitter inhibition, likely at Roundup, appears to be more important than inhibition of sugar GRNs in suppressing proboscis extension.

Mime is a second-order bitter neuron that promotes locomotion and partially counteracts sugar input

Finally, we addressed the role of putative downstream bitter neurons, Scapula (previously identified as a second-order bitter neuron),[8] Mime, and Marge. Mime and Marge receive direct input from bitter GRNs, indicating that they are also second-order bitter neurons. Marge also receives substantial input from Mime, and this input is predicted[38] to be cholinergic and therefore excitatory. We obtained split-Gal4 lines labeling Mime and Marge[31] (Figure S3A; no lines are available for Scapula) and performed calcium imaging. Mime responded to bitter taste delivered to the proboscis, but not to water or sugar, and showed onset and offset responses resembling those in bitter GRNs (Figure 5E).[39] We did not observe bitter responses in Marge (n = 7 flies; Figure S6B), but we cannot rule out potential responses to other bitter compounds or concentrations.

We then tested whether Mime or Marge activation affects taste-related behaviors. Mime activation caused a strong increase in forward velocity (Figure 5F), similar to bitter GRNs.

(D) Summary of behavioral results with sugar interneuron activation. Red, significant and strong phenotype; light red, phenotype that was relatively weak or only significant when analyzing different time points than usual; dashed, phenotype observed with only a subset of lines.
(E and F) Locomotor suppression elicited by activating sugar interneurons in fed versus starved flies (E; n = 4–12 trials) or flies raised on CD versus HSD (F; n = 6–11 trials).
(G) Heatmap showing the number of shared targets for each pair of second-order neurons. Neuron names are colored orange or red if they affected a single behavior or multiple behaviors, respectively.
(H) Schematic of the sugar circuit. Arrow thickness represents connection strength (see STAR Methods for more details).
Flies were 1-day starved in (B), (C), and (F). Numbers in parentheses after the neuron name denote which split-Gal4 line was used. Behavior with medium light intensity is shown except for Zorro (B) and Bract (C), for which high intensity is shown. See also Figures S3 and S4.



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Co-activation of Mime along with sugar GRNs revealed that Mime partially counteracts sugar-evoked stopping but is not sufficient to override the sugar pathway to cause locomotor enhancement (Figure 5F), as bitter GRNs do. Mime activation did not significantly affect positional preference or proboscis extension (Figures S6C and S6D). Activation of Marge did not elicit significant effects on locomotion or preference and did not suppress sugar-evoked stopping, although we noted that Marge unexpectedly elicited modest proboscis extension (Figures S6E–S6G). These experiments uncover a new second-order bitter neuron, Mime, that regulates a subset of bitter-evoked behaviors and likely acts in parallel with other bitter neurons that suppress the sugar pathway.

DISCUSSION

Using sugar taste as a model, we investigated how neural circuits are organized to drive multiple behavioral responses to a single sensory cue. We found that circuits mediating different innate responses to sugar diverge at the level of second- and third-order neurons, whereas circuits for innate and learned behaviors may diverge earlier. Connectome-based simulations revealed synergistic interactions between sugar interneurons and predicted that bitter input suppresses the third and fourth layers of the sugar circuit. Consistent with this prediction, optogenetic experiments suggested that bitter input inhibits third- and fourth-order sugar neurons, whereas hunger and diet modulate earlier nodes of the circuit, revealing distinct sites of modulation.

Behavioral responses to GRN activation

Optogenetic activation of sugar GRNs generally reproduced known effects of sugar, but one counterintuitive result was that innate preference switched to avoidance with higher-intensity stimulation, an observation also noted in a previous study.[40] With natural sugar, positional preference increases with concentration and does not reverse.[12] One possibility is that optogenetically induced avoidance results from unnatural levels of sugar GRN activation that cause compensatory changes. Alternatively, different subsets of *Gr64f-Gal4*-expressing neurons may promote opposing behaviors[41] and strong activation may recruit aversive circuits. Flies naturally avoid sugar in some contexts, such as egg laying, and subsets of sugar GRNs may elicit aversion.[16,17]

Hunger state modulated sugar-evoked behaviors more strongly than bitter-evoked behaviors. This difference is likely adaptive: caloric intake promoted by sweet taste is only necessary when energy stores are low, whereas the role of bitter taste in preventing toxin ingestion is important regardless of energy state. However, our results show that hunger can act selectively to modulate different behaviors elicited by the same taste. For example, when activating bitter GRNs, we found that hunger enhances learned aversion but does not affect locomotion or innate aversion, and previous work showed that hunger suppresses the inhibitory effect of bitter on proboscis extension.[5,42] To achieve this selective modulation, some hunger-dependent changes must occur downstream of sensory neurons. For the sugar pathway, our optogenetic results suggest that hunger modulates locomotor suppression by acting early in the circuit, potentially at GRNs,[20] but modulation of other behaviors may involve downstream neurons.[8,28]

Divergence of neural circuits for different behaviors

Where and how does a sensory circuit diverge into different pathways that mediate different behaviors? If the sensory stimulus is complex, this divergence may occur after several layers of sensory processing. This may be the case for vision.[43] In contrast, a clear divergence emerges early in the olfactory system, as second-order neurons project to multiple brain regions with distinct behavioral roles.[44] Taste is typically considered a relatively simple stimulus, suggesting that different behavioral pathways may diverge early, but surprisingly little is known about this in mammals. The canonical taste pathway relays taste information from the tongue to the gustatory cortex,[45] and different behavioral pathways may diverge there. For example, projections from the gustatory cortex to the amygdala regulate innate licking behavior but are dispensable for associating tastes with learned actions.[46]

We found that circuits mediating different innate responses to sugar in *Drosophila* involve overlapping second- and third-order neurons but distinct fourth-order neurons. The functional diversity of second-order neurons—with some neurons having broader roles than others—may explain why so many second-order cell types are needed. Our results also suggest that circuits for innate versus learned taste behaviors may diverge earlier, at the first synapse. Associative learning requires the mushroom body,[47] located in the superior protocerebrum, so it is perhaps unsurprising that neurons in the proboscis extension circuit—which reside solely in the SEZ—are unable to drive learning. Previous studies have identified second-order sugar neurons projecting to the superior protocerebrum[48] as well as neurons conveying input from the SEZ to the mushroom body,[49] some of which may comprise the circuit for odor-sugar learning.

Figure 4. Computational simulations reveal synergistic roles of sugar interneurons and pathways for bitter suppression

(A and B) Activity of Fdg, Foxglove, and Bluebell in response to activation of second-order neurons (A) or activation of sugar GRNs while silencing second-order neurons (B). The legend below the graphs indicates which neurons were activated (A) or silenced (B).

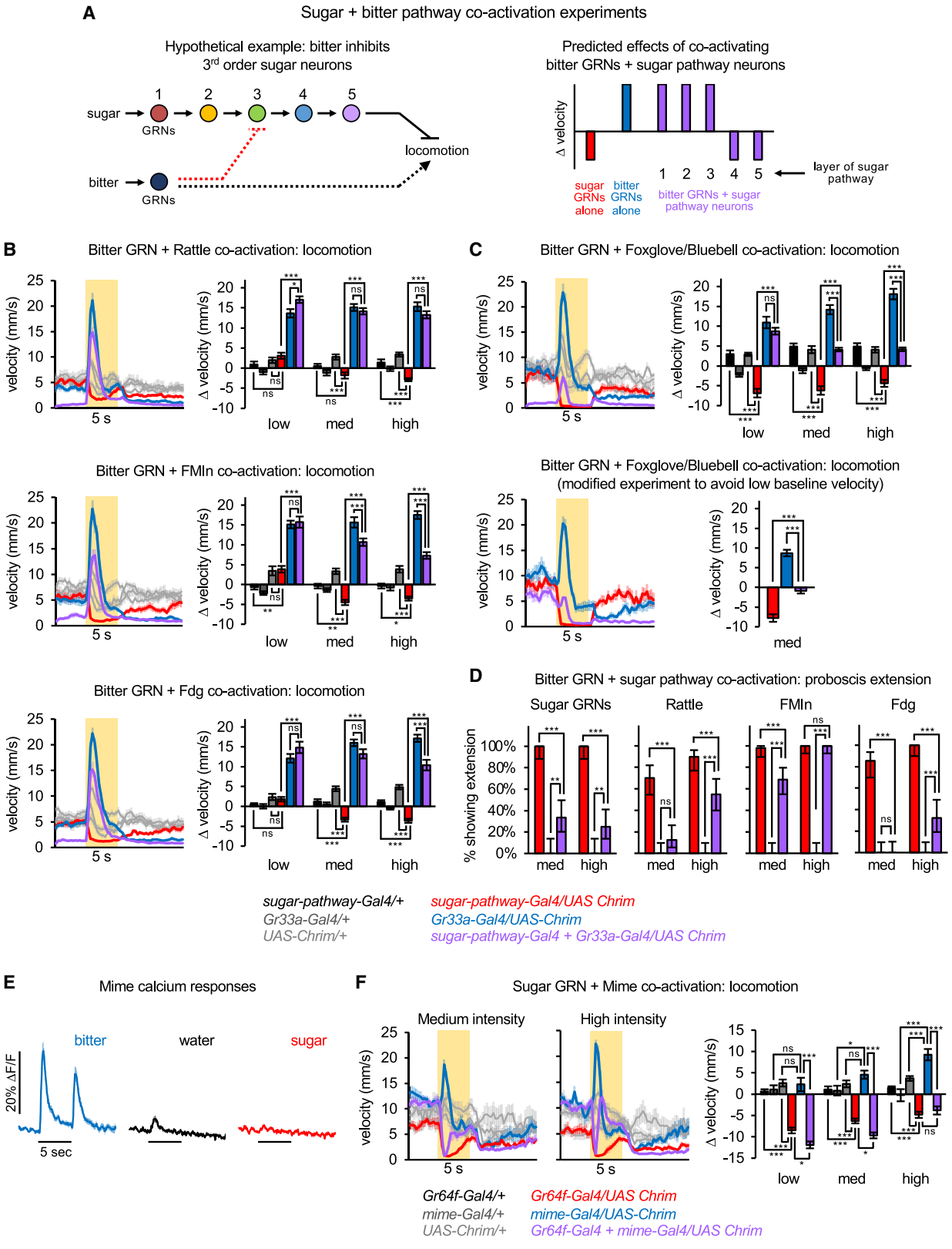
(C and D) Effects of bitter GRN co-activation on all sugar-responsive neurons (C) or known sugar circuit neurons (D). Suppression of some neurons at 50-Hz stimulation was not determined (n.d.) because they were not activated by sugar GRNs alone.

(E and F) Effects on all sugar-responsive neurons (E) or known neurons (F) when sugar GRNs were co-activated with Scapula, Mime, or Marge.

(G–I) Effects on all sugar-responsive neurons (G and H) or known neurons (I) when sugar and bitter GRNs were co-activated while silencing Scapula, Mime, or Marge. In (G), axes represent firing rate with sugar + bitter GRN stimulation normalized to firing rate with sugar stimulation alone. Points above the line of equality represent neurons whose bitter suppression was reduced by neuronal silencing.

(J) Diagram summarizing interactions between the sugar and bitter pathways based on simulations. Flat arrows represent inhibition.

Stimulation intensities used were 200 Hz (A, F, G, and I), 100 Hz (B, bottom), or 50 Hz (B, top and middle; C; E, left). Error bars represent standard deviation. See also Figure S5 and Table S1.



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Interactions between the sugar and bitter pathways

When sugar and bitter GRNs were co-activated, the effects of bitter GRNs dominated. The taste circuit likely evolved this strict hierarchy due to the potentially deadly consequences of ingesting toxins, as preventing toxin ingestion should generally take precedence over caloric consumption. Although bitter input can inhibit sugar GRNs,[22] results from our co-activation experiments suggest that downstream inhibition is critical—with inhibition likely occurring at the third layer to override locomotor stopping and at the fourth layer to suppress proboscis extension.

Our simulations suggest that bitter suppression of the sugar circuit occurs via multiple pathways, including the neurons Scapula, Mime, and Marge, and different neurons inhibit different parts of the sugar circuit. We showed that Mime, a second-order bitter neuron, promotes locomotion but was not sufficient to drive other bitter-evoked behaviors. Thus, the bitter pathway may contain functionally diverse second-order neurons that regulate different behaviors, similar to the sugar pathway.

Limitations of the study

The experiments in this study rely primarily on optogenetic activation, which may not activate neurons at physiological levels. Behavioral phenotypes observed with optogenetic activation demonstrate that the neuron is connected to downstream pathways that mediate the behavior, but we cannot conclude that the neuron mediates the behavior under natural conditions. In addition, this study focuses on known neurons in the sugar circuit for which genetic lines are available; additional neurons may also contribute to sugar-evoked behaviors. Finally, our conclusions regarding the sites where modulatory factors act on the sugar pathway are inferred based on optogenetic manipulations and should be verified using functional imaging or electrophysiological recordings.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Anita Devineni (anita.devineni@emory.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data reported in this paper will be shared upon request to the lead contact.
- Code for whole-brain simulations is posted publicly on GitHub at https://github.com/avdevineni/sugar-circuit.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

A.V.D. conceived and supervised the project, conducted proboscis extension experiments, and performed connectome analyses and model simulations. R.V.J., C.X.W., L.N., P.W., F.V.L.-P., H.A.L., and J.U.D. conducted all other behavioral experiments. T.J.P. conducted calcium imaging experiments. R.V.J., C.X.W., T.J.P., and F.V.L.-P. performed brain dissections and immunostaining to confirm Gal4 expression patterns. A.V.D. generated figures and wrote the manuscript with input from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- EXPERIMENTAL MODEL AND SUBJECT DETAILS
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Figure 5. Co-activation of neurons in the sugar and bitter pathways reveals sites of interaction

(A) Left: schematic of a hypothetical example where bitter input suppresses third-order sugar neurons to override sugar-evoked stopping. Right: predicted effects of co-activating bitter GRNs with sugar pathway neurons at different levels. The graph shows the sugar pathway dominating for co-activation with fourth- and fifth-order neurons, but intermediate effects are also possible.

(B and C) Effects of co-activating bitter GRNs with second- or third-order neurons (B; $n = 9–12$ trials) or fourth-order neurons (C) on locomotion. The co-activation genotype (purple) always showed low baseline velocity, which makes it difficult to compare the magnitude of locomotor changes to controls. In (C), co-activation with Foxglove + Bluebell appeared to cause transient locomotor stimulation followed by locomotor suppression (top; $n = 8$ trials), so we repeated the experiment under conditions where baseline velocity was not impaired (see STAR Methods) and confirmed that locomotor suppression still occurs ($n = 5–8$ trials). Locomotor traces show velocity with medium-intensity light except for Rattle (B), for which high-intensity light is shown.

(D) Effects of co-activating bitter GRNs with sugar pathway neurons on proboscis extension ($n = 24–51$ flies). *Gr33a-Gal4/UAS-Chrim* (middle bars) showed 0% extension in every experiment. Error bars represent 95% confidence intervals calculated by the Wilson/Brown method.

(E) Mime showed a GCaMP response to bitter taste (10 mM denatonium), but not water or sugar (500 mM sucrose) ($n = 29–36$ trials, 6 or 7 flies).

(F) Mime activation (blue) increased locomotion, and co-activating Mime with sugar GRNs (purple) caused flies to initially stop before resuming movement more quickly than flies with sugar GRN activation alone (red) ($n = 6–8$ trials).

Flies were 1-day starved for behavioral experiments. Bar graphs for locomotion show change in velocity at light onset. Fdg line (2) was used for all co-activation experiments. See also Figure S6.

○ Whole-brain simulations
● QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit anti-DsRed	Clontech	RRID:AB_10013483
Alexa Fluor 568 goat anti-rabbit	Life Technologies	RRID:AB_10563566
Chemicals, Peptides, and Recombinant Proteins		
All *trans*-retinal	Sigma-Aldrich	R2500
Sucrose	Sigma-Aldrich	S9378
Denatonium benzoate	Sigma-Aldrich	D5765
Experimental Models: Organisms/Strains		
Drosophila, UAS-Chrimson-TdT^{VK5}	Duistermars et al. (2018)[50]	N/A
Drosophila, UAS-GCaMP6f^{attp40}	Chen et al. (2013)[51]	BDSC: 42747
Drosophila, UAS-Kir2.1	Baines et al. (2001)[32]	BDSC: 6595
Drosophila, lexAop-Chrimson^{attp18}	D. Hattori	N/A
Drosophila, Gr64f-Gal4	Dahanukar et al. (2007)[2]	BDSC: 57669
Drosophila, Gr5a-lexA	Gordon and Scott (2009)[52]	BDSC: 93013
Drosophila, Gr33a-Gal4 (used in Figure 2)	Moon et al. (2009)[53]	BDSC: 31425
Drosophila, Gr33a-Gal4 (used in Figure 5)	Weiss et al. (2011)[54]	BDSC: 57623
Drosophila, zorro-Gal4(1) (*SS67406*)	Shiu et al. (2022)[8]	N/A
Drosophila, zorro-Gal4(2) (*SS67405*)	Shiu et al. (2022)[8]	N/A
Drosophila, clavicle-Gal4 (*SS48947: VT020732-p65ADZp^{attp40}; R17G10-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, G2N-1-Gal4 (*SS56399: R12C04-p65ADZp^{attp40}; VT020839-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, rattle-Gal4 (*SS50091: VT006545-p65ADZp^{attp40}; VT023745-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, usnea-Gal4 (*SS37122: VT037525-p65ADZp^{attp40}; VT033627-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, FMIn-Gal4 (*SS48949: R81E10-p65ADZp^{attp40}; R21H11-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, phantom-Gal4(1) (*SS44879: R20G06-p65ADZp^{attp40}; R82F02-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, phantom-Gal4(2) (*SS44877: R82F02-p65ADZp^{attp40}; R20G06-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, bract-Gal4(1) (*SS31320: R25A01-p65ADZp^{attp40}; VT058723-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, bract-Gal4(2) (*SS31386: R25A01-p65ADZp^{attp40}; R37D11-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, Fdg-Gal4(1) (*SS31333: R81E10-p65ADZp^{attp40}; VT037804-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, Fdg-Gal4(2) (*SS46913: R81E10-p65ADZp^{attp40}; R88C07-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, roundup-Gal4(1) (*SS47744: R23G11-p65ADZp^{attp40}; VT003236-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, roundup-Gal4(2) (*SS47745: R11B11-p65ADZp^{attp40}; VT003236-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A
Drosophila, foxglove+bluebell-Gal4 (*SS31328: VT050223-p65ADZp^{attp40}; VT037804-ZpGDBD^{attp2}*)	Sterne et al. (2021)[31]	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Drosophila, mime-Gal4 (SS47275: R28D10-p65ADZp^{attp40}; VT044856-ZpGDBD^{attp2})	Sterne et al. (2021)[31]	N/A
Drosophila, marge-Gal4 (SS32463: R46E07-p65ADZp^{attp40}; VT037488-ZpGDBD^{attp2})	Sterne et al. (2021)[31]	BDSC: 87350
Software and Algorithms		
MATLAB	Mathworks	https://www.mathworks.com
GraphPad Prism, version 9	GraphPad Software	https://www.graphpad.com/features
Code for brain simulations	this study, based on code from Shiu et al.[37]	https://github.com/avdevineni/sugar-circuit (https://doi.org/10.5281/zenodo.13372368)
Expression database for split-Gal4 lines	Sterne et al. (2021)[31]	https://splitgal4.janelia.org

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Fly strains and maintenance

Flies were reared at 25°C on standard cornmeal or cornmeal/molasses food and maintained in constant darkness to avoid activation of Chrimson. All experiments used mated females, which were collected within a few days after eclosion. Flies were placed on food containing 1 mM all *trans*-retinal for 2–4 days prior to testing. During starvation, flies were housed in vials with 1 mM all *trans*-retinal solution on a piece of Kimwipe. The duration of starvation was 1 day except for experiments with bitter GRN activation in Figures S1H and S1I, which also tested 2-day starvation. The high-sugar diet consisted of standard fly food with 30% sucrose (w/v) added, and flies were fed on a high-sugar diet or control diet (standard fly food) for 6–7 days. Our standard fly food uses the Janelia recipe (0.7% agar, 8.0% cornmeal, 2.0% inactivated yeast, 5.0% molasses, 0.50% propionic acid, 0.20% Tegosept).

Genotypes used for each experiment are specified in the figures. Detailed information regarding fly strains is provided in the key resources table, including full genotypes for split-Gal4 lines if this information was available. For interneuron activation, multiple split-Gal4 lines were available for some neurons. We selected initial lines for testing based on the strength of their proboscis extension phenotype in Shiu et al.[8] and the specificity of their expression pattern, based on images in the Janelia FlyLight Split-Gal4 database (https://splitgal4.janelia.org/) acquired by Sterne et al.[31] We initially tested one split-Gal4 line per neuronal type; if phenotypes were borderline in any behavioral assay we then tested a second line labeling the same neuron (if available). Each neuronal "type" comprises a single neuron per hemisphere with the exception of Bract, which consists of 2 cells, but only one cell appears to be labeled in the split-Gal4 lines.

METHOD DETAILS

Behavioral assays

Assays for locomotion, innate preference, and associative learning were conducted using previously described methods.[30] The behavioral arena contains a circular chamber with a glass cover, and flies are filmed from above using a USB camera. The infrared light for illumination and the red (627 nm) LED array for optogenetic stimulation are located just beneath the chamber. For locomotor and preference assays, low, medium, and high intensity corresponded to 4, 20, and 35 $\mu W/mm^2$. 20–25 flies were tested per trial. Flies were loaded into the chamber using mouth aspiration and given ~3 min to habituate before the experiment started. Flies were filmed at 30 frames/s.

Locomotion and innate preference were assayed sequentially in the same flies. To quantify locomotor effects, light stimulation was presented for 5 s. Each of the three light intensities was delivered to the same flies in increasing order of intensity, with at least 2 min between stimulations to ensure that the behavior had recovered. After an additional rest period, the same flies were then tested for innate preference by delivering low intensity light stimulation to two opposing quadrants for 30 s. The flies then had a 30 s rest period without light, followed by low intensity light stimulation in the other two quadrants for 30 s. Switching the light quadrants ensured that we could assess light preference independently of any spatial bias. This protocol was then repeated sequentially with medium and high intensity light.

For associative learning, the odors used were 3-octanol (1:1000) and 4-methylcyclohexanol (1:750). The flow rate was 100 mL/min. Which odor was used as the CS+ (paired with light) or CS- (not paired with light) was counterbalanced across trials. 4 $\mu W/mm^2$ was initially used for learning experiments with *Gr64f-Gal4*, as this was the intensity that elicited preference in the innate choice assay, but we also tested 30 $\mu W/mm^2$ in Figure S1B. 30 $\mu W/mm^2$ was used to activate *Gr5a*-expressing neurons (Figure S1G) or downstream sugar circuit interneurons (Figure S4C).

All learning assays used 1 min odor-light pairing, following the protocol used in previous learning studies,[55] other than the experiments shown in Figure S1B which used 3 × 10 s pairings. For 1 min pairing, the CS + odor was presented in all quadrants for 1 min

along with light stimulation (thirty 1 s pulses) that started 5 s after odor valve opening. After a 1 min rest, the CS- odor was presented alone for 1 min. Following another 1 min rest, the CS+ and CS- odors were simultaneously delivered to different sets of opposing quadrants for 1 min, allowing the flies to choose between the odors. After another 1 min rest, the two odors were presented again for 1 min but the odor quadrants were switched to control for any spatial bias.

For 10 s odor-light pairing, the CS+ odor was presented for 10 s with light stimulation (three 1 s pulses) that started 10 s after odor valve opening, following the protocol used in a recent study.[41] The CS- odor was presented for 10 s without light. The CS+ and CS- presentations were repeated three times (with CS+ and CS- interspersed) with 2 min between each odor onset. Odor preference tests were performed in the same way as for the 1 min pairing protocol, with flies given 1 min to choose between odors over two separate tests.

Fly videos were analyzed using FlyTracker,[56] and FlyTracker output was analyzed in MATLAB to quantify locomotion or preference. Preference index (PI) values were quantified in 1 s bins. For innate preference, PI was calculated as (# flies in light quadrants – # flies in non-light quadrants)/total # flies. For learning, PI for the conditioned odor was calculated as (# flies in CS+ quadrants – # flies in CS- quadrants)/total # flies. The final PI shown in the figures represents the average PI over the last 5 s of light or odor presentation. Data from the first and second tests were combined, so two trials were obtained for each set of flies.

To analyze locomotion, forward and angular velocities were averaged over 0.33 s bins (10 frames) except for the zoomed-in plots shown in Figures 2A and 2B. (The zoomed-in traces may show higher peak values than the non-zoomed plots for this reason.) To quantify locomotor changes at light onset or throughout the light period, we averaged forward or angular velocity over the first 1 s or entire 5 s of light presentation, respectively, and subtracted these values from the baseline values averaged over the 4 s preceding light onset. Quantifying the light-evoked change in locomotion was more meaningful than quantifying absolute velocity during light because different groups may show different baseline velocities, which could be misinterpreted as an optogenetic effect if the change from baseline is not considered. Differences in baseline velocity complicated the interpretation of some results (Figures 5B and 5C), which is mentioned in the figure legends. For a small number of experiments in which some trials had a very low baseline that made locomotor suppression difficult to assess, we excluded trials with baseline velocity less than 2 mm/s.

For testing co-activation of bitter GRNs with Foxglove and Bluebell in Figure 5C, we noticed that baseline velocity of the co-activation genotype decreased during the course of the experiment, especially if the flies were previously stimulated with light. We therefore we repeated the locomotor assay while only testing flies with medium intensity light (i.e., excluding the low intensity trial that normally comes first), and we additionally excluded trials of any genotype with baseline velocity less than 2 mm/s. This resulted in a dataset where co-activation flies did not show impaired baseline velocity, enabling us to confirm that co-activation causes locomotor suppression.

Although we initially tested the effect of activating each sugar interneuron on locomotion, preference, and learning, for follow-up experiments testing neuronal silencing (Figure S4E), modulation by hunger state (Figure 3E), or modulation by diet (Figure 3F), we focused on analyzing locomotor suppression driven by Rattle, FMIn, or Fdg because these phenotypes represented the most robust behaviors elicited by sugar interneurons.

Proboscis extension assays were conducted in 1-day starved flies as previously described.[33,57] Flies were immobilized on their backs with myristic acid wax, and the two anterior pairs of legs were glued down so that the proboscis was accessible. Flies recovered from gluing for 30–60 min in a humidified chamber and were water-satiated before testing. To activate neurons with Chrimson, a red LED (617 nm) was manually turned on for ~1 s. Each fly was given two chances to respond to the light stimulus. A similar protocol was used to test for light suppression of sugar-evoked proboscis extension except that sucrose and light were delivered simultaneously. At the end of each experiment, flies that did not respond during the experiment were tested with a positive control (500 mM sucrose) and were excluded from analysis if they did not respond.

Immunostaining

Immunostaining experiments were performed as previously described.[30,39] Briefly, brains were dissected in phosphate buffered saline (PBS), fixed for 15–20 min in 4% paraformaldehyde, washed multiple times with PBS containing 0.3% Triton X-100 (PBST), blocked with 5% normal goat serum for 1 h, incubated with primary antibody at 4°C for 2–4 days, washed in PBST, incubated with secondary antibody at 4°C for 1–2 days, washed in PBST and PBS, and mounted in Vectashield. We used rabbit anti-DsRed (1:500) as the primary antibody and Alexa Fluor goat anti-rabbit (1:500) as the secondary antibody. Images were acquired on an Olympus spinning-disk confocal microscope (Clavicle, Rattle, G2N-1, Foxglove+Bluebell, Mime, and Marge) or a Keyence BZ-X810 microscope (all other neurons). Images were processed using FIJI. We did not stain peripheral tissues or the ventral nerve cord, but nerve cord staining for most of the split-Gal4 lines can be viewed at the FlyLight split-Gal4 expression database (splitgal4.janelia.org).

Calcium imaging

Calcium imaging methods were similar to those previously described.[33,39] Briefly, flies were immobilized on clear tape in an imaging chamber and an imaging window was cut on the anterior surface of the head, with the antennae removed. The brain was immersed in modified artificial hemolymph in which 15 mM ribose is substituted for sucrose and trehalose.[13] Images were acquired at ~10 Hz using a two-photon laser scanning microscope (Bruker) with a Ti:S laser (Coherent) with an excitation wavelength of 925 nm and a 20x water-immersion objective. Tastants were delivered to the proboscis using a custom-built solenoid pinch valve system

controlled by MATLAB software, and tastants were removed by a vacuum line controlled by a separate pinch valve. Proper taste delivery was monitored using a side-mounted camera. At least 1 min rest was given between trials to avoid habituation. Calcium imaging data were analyzed using MATLAB code from previous studies.[33,39] The average pixel intensity within a region of interest was calculated for each frame, the average fluorescence over ~2 s prior to taste delivery was used as the baseline signal, and the ΔF/F (change in fluorescence divided by baseline) for each frame was then calculated.

Connectome analysis

Annotated neuronal connectivity from a whole-brain dataset[34,35,58] was downloaded from FlyWire using version 630. Each of the sugar circuit interneurons has been previously identified and labeled in this connectome.[8] To examine the targets and output similarity between second-order neurons, and to make the diagram in Figure 3H, we focused on neurons in a single hemisphere (left). To determine which second-order neurons provide input to Fdg, Bract, Foxglove, or Bluebell, we analyzed inputs to cells in both hemispheres. Cell IDs are: Zorro (720575940629888530), Rattle (720575940638103349), FMIn (720575940614763666), G2N-1 (720575940620874757), Clavicle (720575940655014049), Phantom (720575940616103218), Usnea (720575940632648612), Fdg (720575940631997032, 720575940647030324), Bract (720575940645045527, 720575940610001220, 720575940626557442, 720575940627285267), Foxglove (720575940611160930, 720575940628755815) and Bluebell (720575940631168057, 720575940635166571). To quantify the output similarity between two neurons we used cosine similarity, a metric that has been previously used in connectomic analyses.[59] We calculated the cosine of the angle between the vectors containing the number of synapses onto each output neuron. These vectors are n-dimensional, where n is the total number of output neurons for the two input neurons.

For the diagram in Figure 3H, arrow thickness represents connection strength: thin = 5-9 synapses; medium = 10–19 synapses; thick = 20+ synapses. For simplicity, the diagram omits predicted inhibitory connections (from Phantom and Usnea), feedback connections from 3rd to 2nd order neurons, and weak connections from FMIn to Foxglove and Rattle to Bluebell that are only observed in one hemisphere.

Whole-brain simulations

Brain simulations were conducted in Python based on code provided publicly by Philip Shiu (https://github.com/philshiu/Drosophila_brain_model), and details about the model are described in Shiu et al. (2023).[37] Code for all simulations in this paper can be found on GitHub (https://github.com/avdevineni/sugar-circuit). Simulations were based on connectivity data in version 630 of the Flywire connectome. We used the default parameters for the model, which were chosen by Shiu et al.[37] based on published experimental data and include: −52 mV resting potential, −52 mV reset potential after a spike, −45 mV spiking threshold, 2.2 ms refractory period, 5 ms tau for synapse decay, 1.8 ms time delay from spike to change in membrane potential, and 0.275 mV synaptic weight. In all simulations, neurons were stimulated for 1 s and activity of neurons in the left hemisphere was quantified. Neurons were activated or silenced bilaterally with the exception of GRNs, because only sugar and bitter GRNs on the left side have been thoroughly annotated and we did not want to risk activating incorrect neurons. All simulations were repeated over 30 trials, and mean values were quantified. Simulations were performed with multiple stimulation intensities (typically 50, 100, and 200 Hz), corresponding to a similar range as tested in Shiu et al.[37] Results from the most informative intensity are shown in Figure 4, and additional results are shown in Figure S5 or reported in Table S1. In Figures 4A and 4B, we tested activation or silencing of every individual second-order neuron and every pair of neurons; for combinations of 3 or more neurons, we focused on only testing the excitatory neurons.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed using GraphPad Prism, Version 9. All graphs represent mean ± SEM except for bar graphs showing the results of simulations (Figures 4A, 4B, and S5A), which represent standard deviation, and bar graphs showing results with proboscis extension (Figures 5D, S3B, S6D, and S6G), which represent 95% confidence intervals calculated by the Wilson/Brown method.

Sample sizes are listed in the figure legends. For statistical analyses of locomotion, preference, and learning data, each trial was considered to be a single data point (“n”). For locomotion assays, each trial for a given light intensity represents a different set of flies. For preference and learning assays, two trials were obtained from each set of flies, representing trials in which light or odor stimuli were presented in opposing quadrants (see Figures 1D and 1H).

For most experiments, two groups were compared using unpaired t-tests, and more than two groups were analyzed using one-way ANOVA followed by Dunnett’s test comparing experimental flies to each control. Analyses using other statistical tests are listed here.

- Figures 2H–2J and S2A–S2C: two-way ANOVA followed by Dunnett’s post-tests comparing experimental flies to each control or Bonferroni’s post-tests comparing CD and HSD.
- Figures 4A, 4B, 5B, 5C, 5F, S5A, S6A, and S6E: One-way ANOVA followed by Bonferroni’s post-tests. Note that in Figures 4A and 4B, statistically significant comparisons showing synergistic interactions between second-order neurons are depicted in the figure; not all significant results are shown and not all possible comparisons were performed.
- Figures 5D, S3B, S6D, and S6G: Fisher’s exact test.

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Supplemental information

Overlap and divergence of neural circuits mediating distinct behavioral responses to sugar

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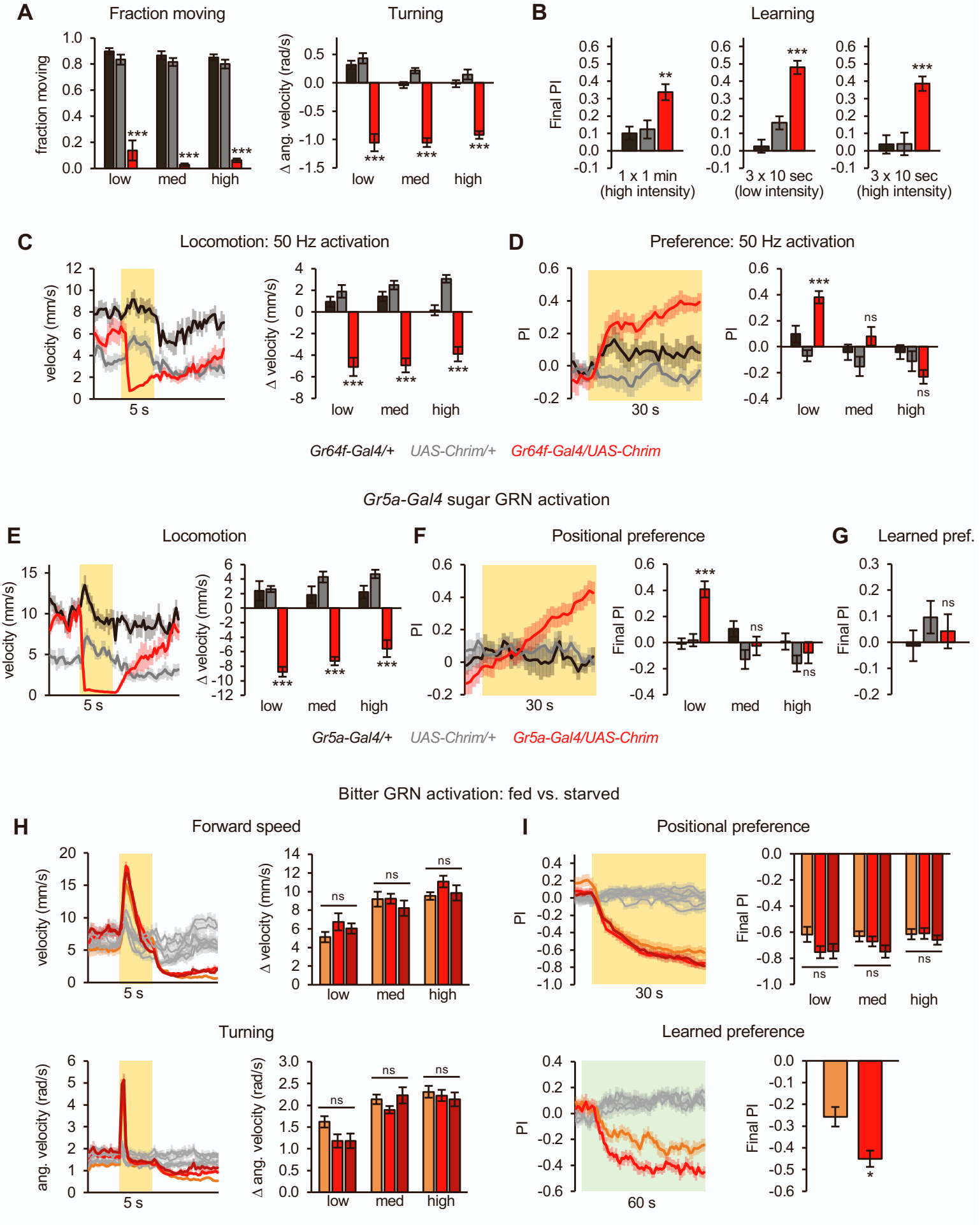

A
Fraction moving
fraction moving
low
med
high
Turning
Δ ang. velocity (rad/s)
B
Learning
Final PI
1 x 1 min
(high intensity)
3 x 10 sec
(low intensity)
3 x 10 sec
(high intensity)
C
Locomotion: 50 Hz activation
velocity (mm/s)
5 s
Δ velocity (mm/s)
D
Preference: 50 Hz activation
PI
30 s
Gr64f-Gal4/+
UAS-Chrim/+
Gr64f-Gal4/UAS-Chrim
Gr5a-Gal4 sugar GRN activation
E
Locomotion
F
Positional preference
G
Learned pref.
Gr5a-Gal4/+
UAS-Chrim/+
Gr5a-Gal4/UAS-Chrim
Bitter GRN activation: fed vs. starved
H
Forward speed
I
Positional preference
Turning
ang. velocity (rad/s)
Learned preference
60 s
Gal4/+ and UAS/+ controls
Gr33a-Gal4/UAS-Chrim: fed vs. 1d or 2d starved

Figure S1, related to Figure 1. Additional characterization of behaviors elicited by sugar GRN activation and comparison to bitter GRN activation

(A) Sugar GRN activation in fed flies decreased the fraction of flies moving (left) and turning speed (right) (n = 12 trials). Values are quantified for the first 1 sec of light stimulation.

(B) Left: Pairing odor with activation of sugar GRNs at a higher light intensity (30 μW/mm^2) than in Figure 1 (4 μW/mm^2) elicited learned odor preference (n = 24 trials, 12 sets of flies). Middle and right: Pairing odor and sugar GRN activation 3 times for 10 sec each at two different light intensities (4 or 30 μW/mm^2) elicited similar learned preference (n = 12-28 trials; 6-14 sets of flies) as one pairing for 60 sec (left graph and Figure 1I-K). All three experiments were conducted with one-day starved flies.

(C-D) Because the locomotion and preference assays in Figure 1 were performed with continuous light stimulation, we also tested 50 Hz pulsed light stimulation. Activating sugar GRNs using 50 Hz pulsed light in one-day starved flies elicited similar effects on locomotion (C; n = 8-9 trials) and positional preference (D; n = 12-18 trials, 6-9 sets of flies) as those observed with continuous light (Figure 1C, 1E-F).

(E-G) Activation of sugar GRNs in starved flies using *Gr5a-Gal4* elicited locomotor suppression (E; n = 8 trials) and positional preference (F; n = 16 trials, 8 sets of flies) but not learned preference (G; n = 24 trials, 12 sets of flies). The absence of learning could be due to the fact that *Gr64f-Gal4* labels more sugar GRNs across the body than *Gr5a-Gal4* [S1-2], but it is also possible that stronger pairing protocols could elicit learning.

(H-I) Activation of bitter GRNs using *Gr33a-Gal4* elicited aversive behaviors that are similar in fed and starved flies, with the exception of stronger aversive learning in starved flies. Data for fed flies are the same data shown in Deere et al., 2023 [S3]. (H) Locomotor effects elicited by bitter GRN activation in fed and starved flies. Traces show forward and angular velocity for medium intensity light stimulation (n = 10-12 trials). Bar graphs quantify the change in these parameters at light onset. (I) Innate positional aversion (top; n = 20-24 trials, 10-12 sets of flies) and learned aversion (bottom; n = 22-24 trials, 11-12 sets of flies) elicited by bitter GRN activation in fed and starved flies.

Unless otherwise specified, for all supplemental figures, two groups were compared using unpaired t-tests and more than two groups were analyzed using one-way ANOVA followed by Dunnett's test comparing experimental flies to each control. See STAR Methods for statistical tests used.

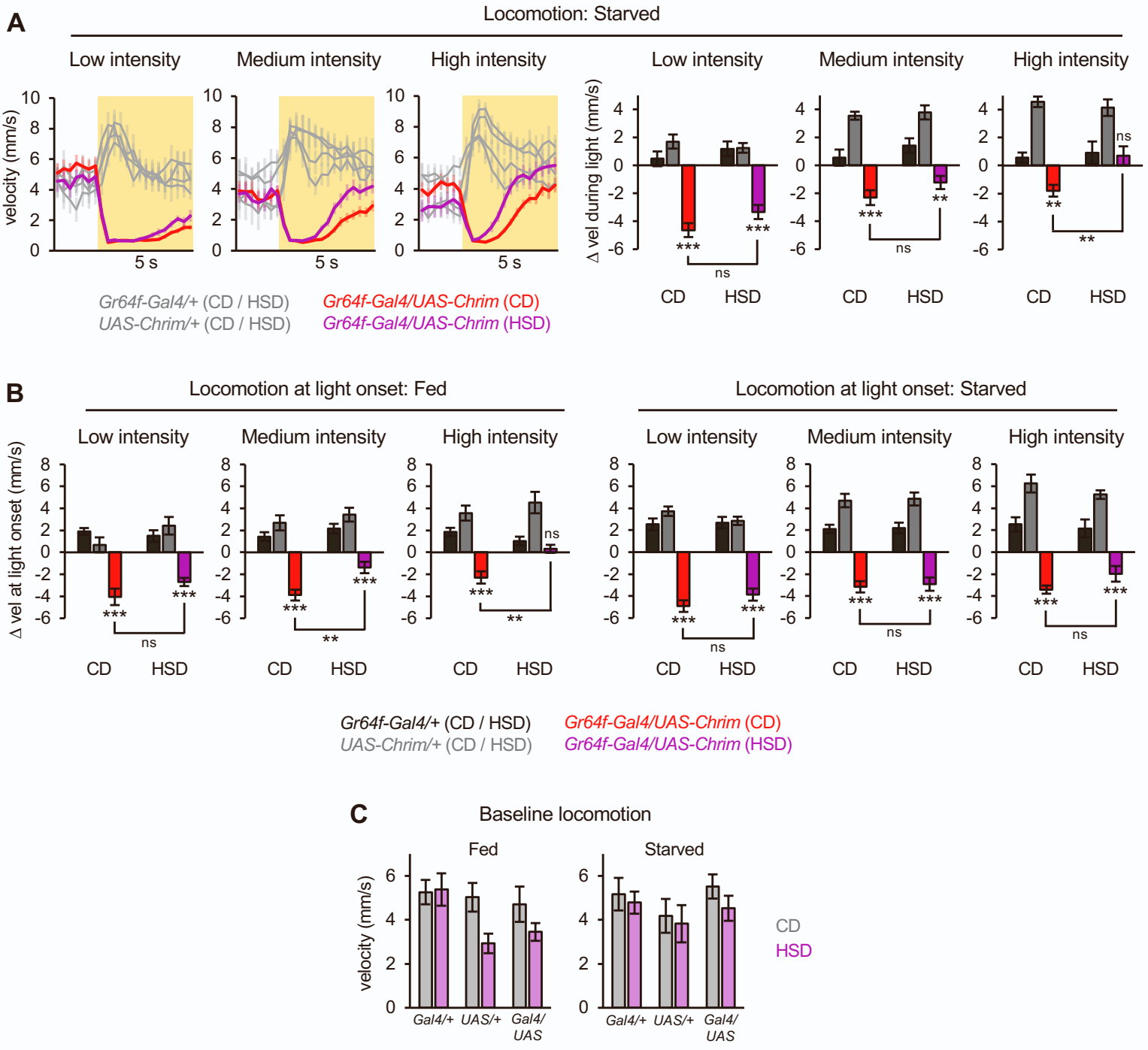


Figure S2, related to Figure 2. Additional characterization of behavioral responses in flies on a high-sugar diet

(A) Locomotor suppression elicited by sugar GRN activation in one-day starved flies was more transient for flies on a high-sugar diet (HSD) compared to a control diet (CD) (n = 7-10 trials), similar to the effect in fed flies (Figure 2H). Bar graphs show change in velocity during the entire light period. The change in velocity at light onset is shown in Figure S2B.

(B) Change in velocity at light onset (first 1 sec) quantified from the data in Figure 2H and S2A (n = 7-10 trials).

(C) Baseline locomotion for experiments shown in Figure 2H and S2A (n = 7-10 trials). In fed flies, there was a significant overall effect of diet based on two-way ANOVA ($p<0.05$) but no significant effect for any individual genotype based on Bonferroni's post-tests. There was no significant effect of diet in starved flies ($p>0.05$).

Statistics in this figure represent the results of two-way ANOVA followed by Dunnett's post-tests comparing experimental flies to each control or Bonferroni's post-tests comparing CD and HSD.

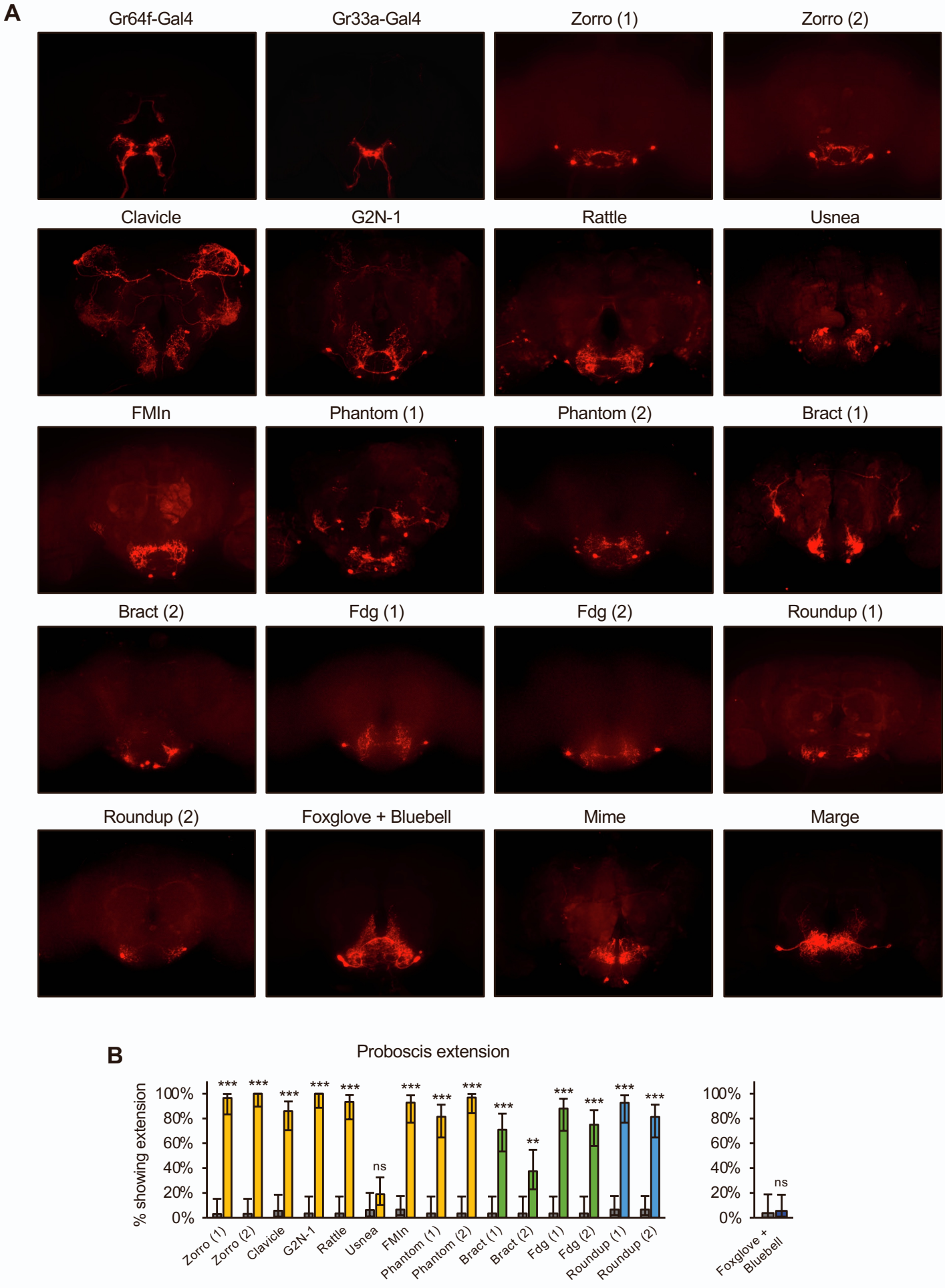

A
Gr64f-Gal4
Gr33a-Gal4
Zorro (1)
Zorro (2)
Clavicle
G2N-1
Rattle
Usnea
FMIn
Phantom (1)
Phantom (2)
Bract (1)
Bract (2)
Fdg (1)
Fdg (2)
Roundup (1)
Roundup (2)
Foxglove + Bluebell
Mime
Marge
B
Proboscis extension
% showing extension
100%
80%
60%
40%
20%
0%
***
**
ns
Zorro (1)
Zorro (2)
Clavicle
G2N-1
Rattle
Usnea
FMIn
Phantom (1)
Phantom (2)
Bract (1)
Bract (2)
Fdg (1)
Fdg (2)
Roundup (1)
Roundup (2)
Foxglove + Bluebell

Figure S3, related to Figure 3. Expression patterns of Gal4 lines and validation of proboscis extension

(A) Expression patterns for Gal4 or split-Gal4 lines used in this study, based on staining for UAS-Chrim-Tdt. Higher-resolution images of split-Gal4 expression taken by the Janelia FlyLight project are available at the Flylight Split-Gal4 database.

(B) Proboscis extension elicited by activating each sugar interneuron in one-day starved flies (Fisher's exact test, n = 25-47 flies). Grey bars represent the *UAS/+* control and colored bars represent experimental flies carrying *split-Gal4* and *UAS-Chrim*; colors represent which layer of the circuit the interneuron resides in (see Figure 3H). Numbers in parentheses after the neuron name denote which split-Gal4 line was used. Left: Activation of interneurons that were previously shown to cause proboscis extension by Shiu et al. [S4] elicited significant proboscis extension for all neurons except Usnea. Note that in the study by Shiu et al., Usnea activation caused proboscis extension only at very high light intensities. Right: Co-activation of Foxglove and Bluebell, fourth-order neurons in the stopping circuit, did not elicit proboscis extension.

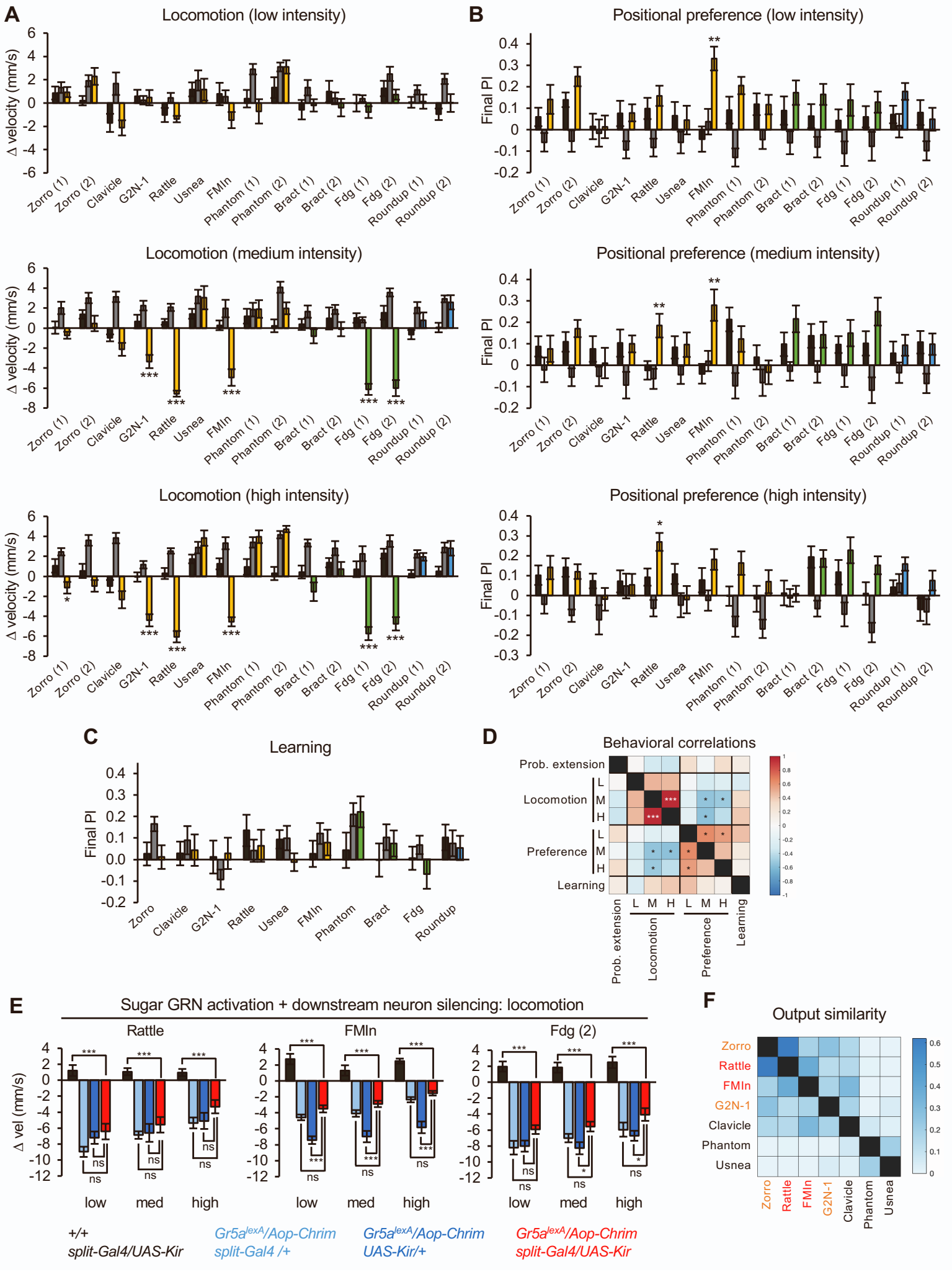

A
Locomotion (low intensity)
Δ velocity (mm/s)
Locomotion (medium intensity)
Locomotion (high intensity)
B
Positional preference (low intensity)
Final PI
Positional preference (medium intensity)
Positional preference (high intensity)
Zorro (1)
Zorro (2)
Clavicle
G2N-1
Rattle
Usnea
FMIn
Phantom (1)
Phantom (2)
Bract (1)
Bract (2)
Fdg (1)
Fdg (2)
Roundup (1)
Roundup (2)
C
Learning
Zorro
Clavicle
G2N-1
Rattle
Usnea
FMIn
Phantom
Bract
Fdg
Roundup
D
Behavioral correlations
Prob. extension
Locomotion
Preference
Learning
L
M
H
E
Sugar GRN activation + downstream neuron silencing: locomotion
Rattle
FMIn
Fdg (2)
Δ vel (mm/s)
low
med
high
+/+
split-Gal4/UAS-Kir
Gr5a^lexA/Aop-Chrim
split-Gal4 /+
Gr5a^lexA/Aop-Chrim
UAS-Kir/+
Gr5a^lexA/Aop-Chrim
split-Gal4/UAS-Kir
F
Output similarity
Zorro
Rattle
FMIn
G2N-1
Clavicle
Phantom
Usnea

Figure S4, related to Figure 3. Additional characterization of behavioral roles of sugar interneurons

(A) Change in forward speed at light onset elicited by activation of each sugar interneuron at each light intensity (n = 7-12 trials). Both Zorro lines caused very brief locomotor suppression (see Figure 3B) at medium and high intensity that was not always significant when quantifying velocity over the first 1 sec, but was highly significant when quantifying velocity over the first 0.33 sec ($p<0.001$ for both lines at medium and high intensities). Locomotor suppression elicited by Zorro was also stronger than it appears in Figure 3B because the traces show binned velocity over 0.33 sec bins.

(B) Innate positional preference elicited by activation of each sugar interneuron at each light intensity (n = 14-24 trials, 7-12 sets of flies). Bar graphs show final PI. Some lines whose final PI was not significant caused statistically significant preference at earlier time points, including Fdg (1) (significant preference with high intensity light at ~13-17 sec after light onset), Fdg (2) (most time points from 11-24 sec after light onset; see Figure 3C), and Bract (1) (all time points from 9-15 sec after light onset; see Figure 3C), as quantified by two-way ANOVA followed by Dunnett's post-tests.

(C) Learned odor preference elicited by activation of each type of sugar circuit interneuron (n = 16-24 trials, 8-12 sets of flies).

(D) Given the overlap between neurons that affected locomotion and innate preference, we asked whether the effects of interneuron activation on different behaviors were correlated. The heatmap shows correlation values (Pearson's r; asterisks denote significance) between behavioral effects elicited by interneuron activation, including average data from all experimental genotypes tested in Figure S4A-C (n = 10 for correlations with learning and n = 15 for all other correlations). L, M, or H denote low, medium, or high intensity light, respectively. Values for "locomotion" represent the change in forward speed at light onset, so negative values represent locomotor suppression. Locomotor suppression and positional preference were modestly correlated at some light intensities, suggesting that these behaviors are mediated by overlapping circuits (and may also be intrinsically related, even though they are dissociable), whereas no other pair of behaviors showed a significant correlation.

(E) Change in forward speed at light onset elicited by activation of sugar GRNs (*Gr5a-lexA* driving *Aop-Chrim*) while silencing downstream sugar interneurons using Kir2.1 (n = 8-12 trials). For all three interneurons, experimental flies with both GRN activation and downstream silencing (red bars) did not significantly differ from both of the controls with GRN activation only (blue bars), indicating that neuronal silencing did not impair locomotor suppression.

(F) Analyzing second- to third-order connectivity using output similarity, which takes connection strength into account, revealed the same two subcircuits as analyzing the number of shared targets in Figure 3G. The heatmap shows cosine similarity of outputs for each pair of second-order neurons. Cosine similarity ranges from 0 to 1, with higher values representing more similar outputs. Neurons are colored orange or red if they affected a single behavior or multiple behaviors, respectively.

Flies were one-day starved for all behavioral experiments. In panels A-C, black and grey bars represent the *Gal4/+* and *UAS/+* controls, respectively, and colored bars represent experimental flies carrying *split-Gal4* and *UAS-Chrim*; colors represent which layer of the circuit the interneuron resides in (see Figure 3H). Second-order neurons are listed in descending order of the number of synapses they receive from sugar GRNs (as reported in Shiu et al., 2022 [S4]). Numbers in parentheses after the neuron name denote which split-Gal4 line was used.

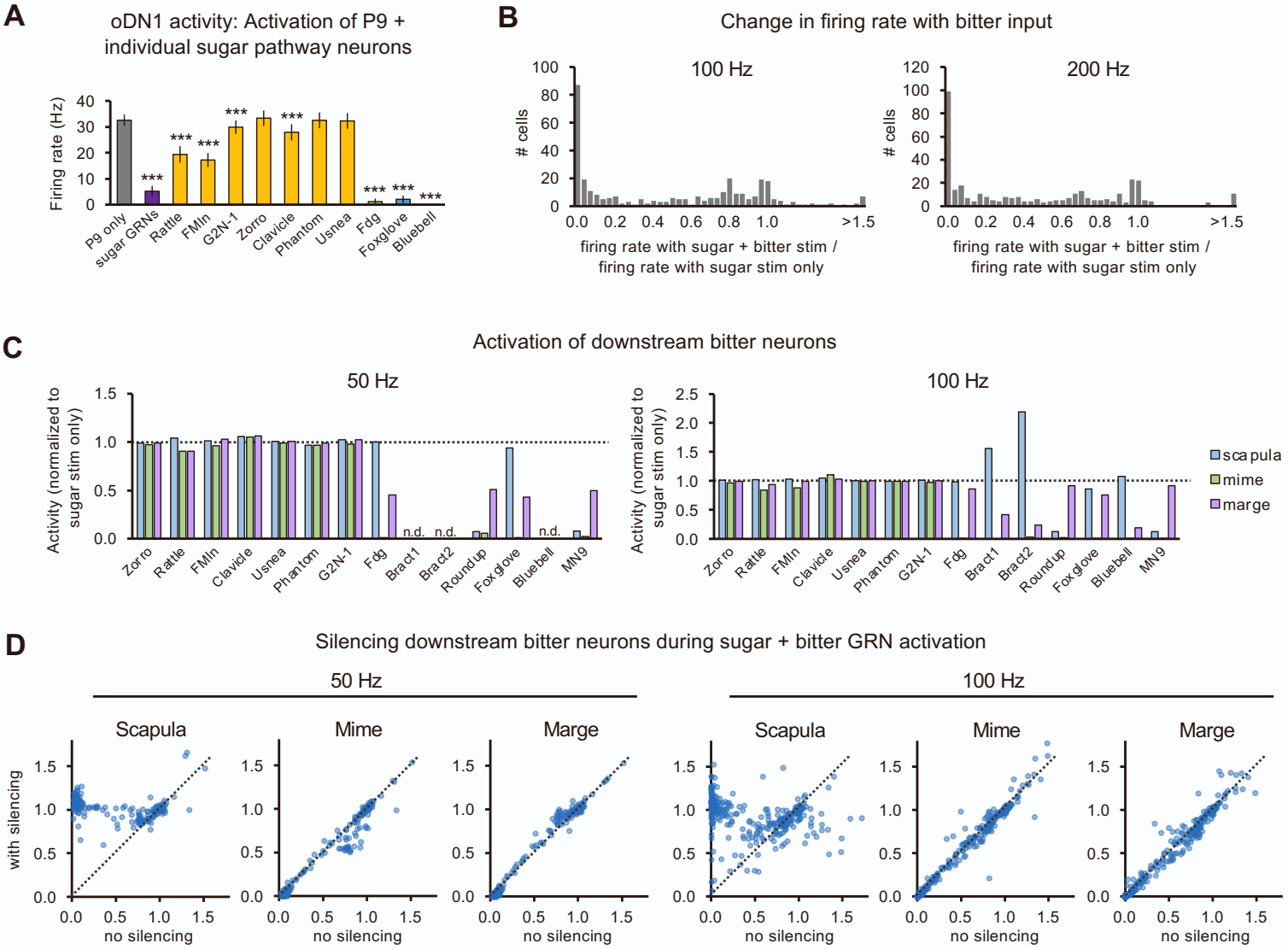


Figure S5, related to Figure 4. Additional results from computational simulations of sugar circuit activity

(A) We simulated the activation of sugar interneurons along with P9, a neuron that promotes walking, and quantified activity in downstream targets of P9 identified in Sapkal et al. [S5]. Neurons were activated at 200 Hz. Of six known P9 targets, only oDN1 was strongly suppressed by activation of sugar GRNs; activity of the other neurons is not shown here. oDN1 activity was strongly suppressed by activating Foxglove, Bluebell, or Fdg, and was moderately suppressed by activating Rattle or FMIn.

(B) Sugar and bitter GRNs were co-activated at 100 Hz (left) or 200 Hz (right). Histograms show the effect of bitter GRN activation on the activity of sugar-responsive neurons. Activation at 50 Hz is shown in Figure 4C.

(C) Sugar GRNs were co-activated with putative downstream bitter neurons Scapula, Mime, or Marge at 50 Hz (left) or 100 Hz (right). Activation of Scapula, Mime, or Marge suppressed the sugar-evoked activation of specific third-, fourth-, or fifth-order sugar neurons, but not second-order neurons. Suppression of some neurons could not be determined (n.d.) at 50 Hz stimulation because they were activated too weakly by sugar GRNs alone. Activation at 200 Hz is shown in Figure 4F.

(D) Sugar and bitter GRNs were co-activated at 50 Hz (left) or 100 Hz (right) while putative downstream bitter neurons were silenced. Silencing Scapula caused a strong reduction in bitter suppression of sugar responses, while Mime and Marge had little to no effect. Axes represent the firing rate with sugar + bitter GRN stimulation normalized to the firing rate with sugar stimulation alone, with the x- and y-axes showing normalized firing with and without downstream bitter neuron silencing, respectively. Points above the line of equality represent neurons whose bitter suppression was reduced by neuronal silencing. Co-activation at 200 Hz is shown in Figure 4G.

All simulations were repeated over 30 trials and error bars show the standard deviation. Statistics in panel A represent one-way ANOVA followed by Bonferroni's post-tests.

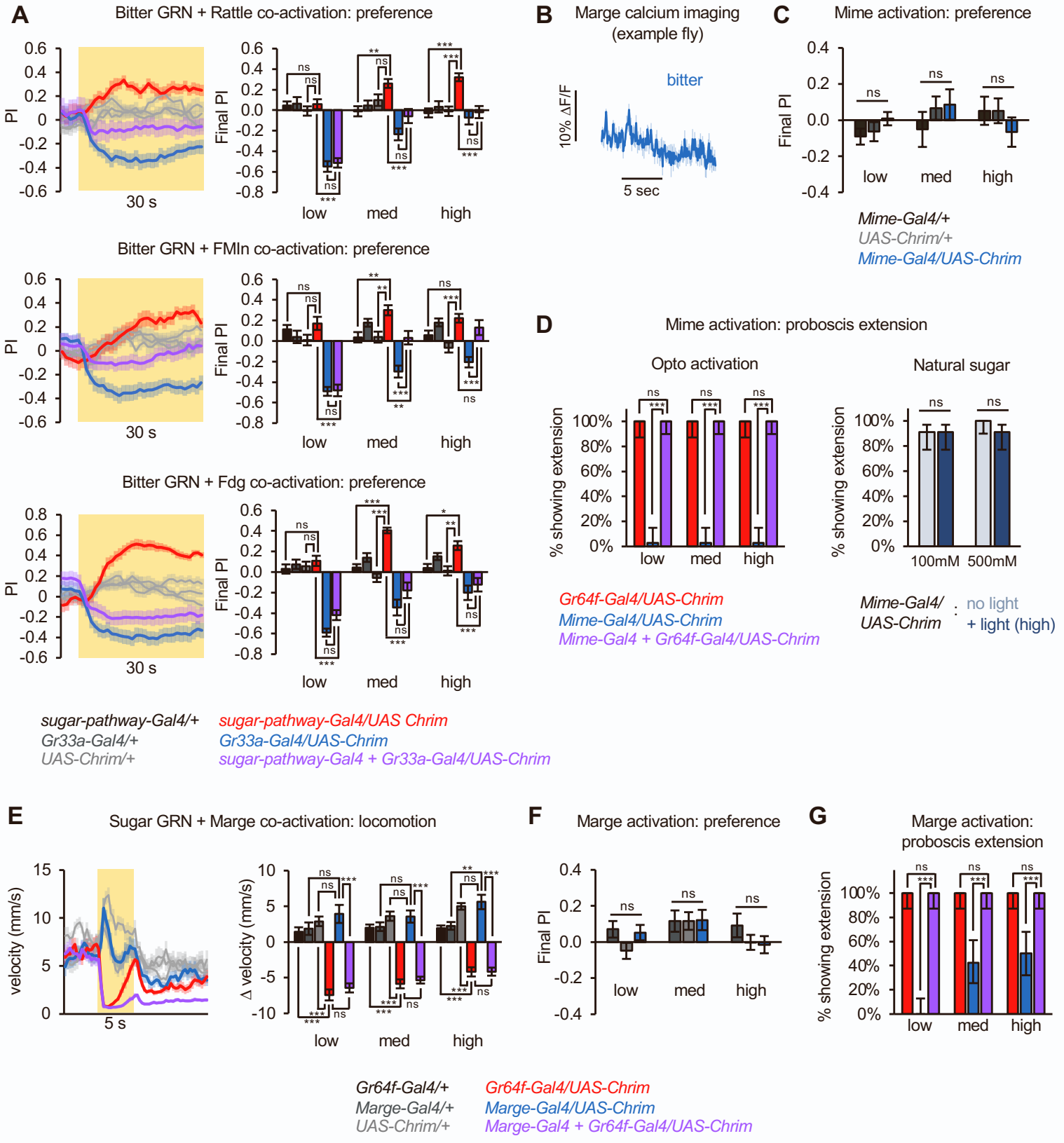

A
Bitter GRN + Rattle co-activation: preference
PI
30 s
Final PI
low
med
high
Bitter GRN + FMIn co-activation: preference
Bitter GRN + Fdg co-activation: preference
sugar-pathway-Gal4/+
Gr33a-Gal4/+
UAS-Chrim/+
sugar-pathway-Gal4/UAS Chrim
Gr33a-Gal4/UAS-Chrim
sugar-pathway-Gal4 + Gr33a-Gal4/UAS-Chrim
B
Marge calcium imaging
(example fly)
bitter
10% ΔF/F
5 sec
C
Mime activation: preference
Mime-Gal4/+
UAS-Chrim/+
Mime-Gal4/UAS-Chrim
D
Mime activation: proboscis extension
Opto activation
Natural sugar
% showing extension
100mM
500mM
Gr64f-Gal4/UAS-Chrim
Mime-Gal4/UAS-Chrim
Mime-Gal4 + Gr64f-Gal4/UAS-Chrim
Mime-Gal4/
UAS-Chrim
: no light
+ light (high)
E
Sugar GRN + Marge co-activation: locomotion
velocity (mm/s)
5 s
Δ velocity (mm/s)
F
Marge activation: preference
G
Marge activation:
proboscis extension
Gr64f-Gal4/+
Marge-Gal4/+
UAS-Chrim/+
Gr64f-Gal4/UAS-Chrim
Marge-Gal4/UAS-Chrim
Marge-Gal4 + Gr64f-Gal4/UAS-Chrim

Figure S6, related to Figure 5. Further characterization of phenotypes with activation of neurons in the sugar and bitter pathways

(A) Co-activating bitter GRNs with second- or third-order sugar neurons caused either strong aversion or intermediate effects on positional preference, depending on the neuron and stimulation intensity (n = 19-24 trials, 10-12 sets of flies). Traces show preference with medium intensity stimulation. However, reduced aversion in the co-activation genotype should be interpreted cautiously because these flies showed very low baseline velocity (see Figure 5B), which may prevent the flies from readily moving between the light and non-light quadrants.

(B) Marge did not show any clear GCaMP responses to bitter taste (10 mM denatonium) delivered to the proboscis. Trace shows the average response for one example fly (n = 6 trials) out of 7 flies imaged.

(C) Activation of Mime did not elicit positional preference or aversion (n = 12-16 trials, 6-8 sets of flies).

(D) Activation of Mime did not cause proboscis extension and did not significantly suppress proboscis extension evoked by sugar GRN activation (left; n = 26-34 flies) or natural sugar (right; n = 34 flies). We anecdotally observed that PER evoked by natural sugar may be less vigorous with light stimulation, but this was not quantified.

(E) Activation of Marge did not significantly affect locomotion (blue) or impair the ability of sugar GRN activation to cause locomotor suppression; in fact, locomotor suppression seemed to last longer with Marge co-activation (n = 8-10 trials). Velocity traces for medium intensity stimulation are shown, and bar graph shows change in velocity at light onset.

(F) Activation of Marge did not elicit positional preference or aversion (n = 20 trials, 10 sets of flies).

(G) Activation of Marge caused moderate proboscis extension at medium and high intensities and did not suppress proboscis extension evoked by sugar GRN activation (n = 26 flies).

Flies were one-day starved for all behavioral experiments. Data in panels A and E were analyzed by one-way ANOVA followed by Bonferroni's post-tests, and data in panels D and G were analyzed by Fisher's exact test. Fdg line (2) was used for co-activation experiments.

Supplemental References

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